



DEGREE PROJECT IN TECHNOLOGY,
FIRST CYCLE, 15 CREDITS
STOCKHOLM, SWEDEN 2019

Prospects of Renewable Energy for the New City of El Alamein, Egypt

An Energy System Model using OSeMOSYS
to obtain the most cost-efficient electricity
production mix

A Minor Field Study

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Abstract

With the motivation to mitigate the effects caused by one of humankind's biggest challenges, climate change, the purpose of this minor field study was to examine the prospects of renewable energy technologies as part of a larger ambition to offer clean and affordable energy for all, in line with United Nations Sustainable Development Goals. The study was conducted for El Alamein in Egypt, a city under construction that will house four million residents. By combining a field study and an interview with literature search, the information needed for an energy system model was gathered. The modelling system OSeMOSYS was thereafter used to calculate the most cost-efficient electricity mix for the model period of 2020 to 2040, based on different scenarios and technologies. The total discounted cost and amount of emissions were thereafter compared between the scenarios, and the most cost-efficient scenario at reducing emissions was identified.

Of the scenarios that were compared to the reference case, the one where 50 % of the electricity was produced from renewables in 2040 proved to be the most cost-efficient option. In addition, this scenario was also the most cost efficient at reducing emissions by a large margin. Regarding the renewable technologies, solar photovoltaics, which represented a majority of the renewable electricity production, was evidently the most cost-efficient technology as well as the one with the highest potential for future implementation, followed by onshore wind power. Moreover, concentrating solar power and waste to energy were proven to have a lower potential. The conclusions drawn were that the city of El Alamein could get a substantial part of its electricity from renewables and that solar PV was the best technology for that purpose. The final conclusion was that there might exist great potential for renewable energy in Egypt.

Key words: Renewable energy, energy system modelling, OSeMOSYS, El Alamein, Egypt

Sammanfattning

Med ambitionen att försöka lindra konsekvenserna av vad som idag är en av människans största utmaningar, klimatförändringen, är syftet med denna minor field study att utvärdera förutsättningarna för förnyelsebara energitekniker i Egypten. Detta ligger i linje med en större ambition att erbjuda ren och prisvärd energi i enlighet med de globala hållbarhetsmålen. Fältstudien utfördes i Egypten och omfattade en ny stad under konstruktion, El Alamein, som förväntas hushålla fyra miljoner invånare. En fältstudie och tillhörande intervju kombinerades med en litteraturundersökning för att erhålla nödvändig information som sedan användes i modelleringen av energisystemet. Modelleringsverktyget OSeMOSYS användes för att erhålla den mest kostnadseffektiva energimixen för åren 2020 till 2040, baserat på ett flertal olika scenarier och tekniker. Den totala diskonterade kostnaden samt mängden utsläpp jämfördes mellan de olika scenarierna, och det mest kostnadseffektiva scenariot för att minska utsläpp identifierades.

Av de scenarier som jämfördes med referensfallet, så var scenariot där förnyelsebara energitekniker stod för 50 % av elproduktionen år 2040 den mest kostnadseffektiva energimixen för att tillfredsställa behovet samt för att minska mängden utsläpp. Av de förnyelsebara energiteknikerna så visade sig solceller, som stod för en majoritet av den förnyelsebara produktionen, vara den mest kostnadseffektiva tekniken då den i samtliga förnyelsebara scenarier prioriterades av modellen. Därmed visade sig den vara den tekniken med högst potential för framtida implementationer, följt av landbaserad vindkraft. Koncentrerad solkraft i form av soltorn visade sig inte vara kostnadseffektiv då den inte bidrog till produktionen i något scenario. Avfallsförbränningen, som bidrog minimalt till energimixen, saknade tillgång till den mängd bränsle som hade krävts för att den skulle kunnat stå för en större andel av produktionen. Därmed konstaterades det att båda dessa teknologier hade en låg potential för framtida implementationer.

De slutsatser som drogs var att staden El Alamein kunde få en betydande del av sin elektricitet från förnyelsebara energikällor samt att solceller var mest lämpad för det syftet då den var mest kostnadseffektiv. Utöver dessa så drogs slutsatsen att det kan finnas stor potential för förnyelsebar energi i Egypten.

Nyckelord: Förnyelsebar energi, energisystemmodellering, OSeMOSYS, El Alamein, Egypten

Acknowledgement

This project would not have been possible without the expertise and guidance from our supervisor Prof. Andrew Martin and our external supervisor Dr. Norhan Ibrahim Ghoneim from Pharos University. Prof. Martin laid the foundation for this bachelor thesis by helping us define the scope of the study and has assisted us throughout the course of this project. Dr. Ghoneim helped us during our whole stay in Alexandria, providing us with the necessary information by organizing field visits, interviews and other practical activities. Dr. Ghoneim has also acted as a link between us and the rest of Pharos University. They have our deepest gratitude for enabling this bachelor thesis and collaboration between KTH and Pharos University.

Special thanks to our Egyptian colleagues Mohammed Khairy, Abdelghfar Hassan, Ahmed S. Elhdad and Ahmed M. Rezk for participating in the collaboration. More specifically, we would like to express our gratitude for their part in the common effort of identifying the technologies and in the gathering of information. Additionally, we would like to thank them for acting as our interpreters during interviews, as well as contributing to making our visit more than just a professional cooperation by showing us the local cultures of Alexandria.

We would also like to show our appreciation for the immense work and support provided by Pharos University. From the day of arrival to the day of departure, constant help was provided, making our stay wonderful. Special recognition to the Dean of the faculty of Engineering, Prof. Mohamed G. Abou-Ali for the warm welcome and for showing great enthusiasm for our project. International coordinator Dr. Menna Gowayed and Amr Mamdouh deserves our special thanks for taking care of day-to-day matters in a splendid fashion.

Least but not last, we would like to thank the course examiner Monika Olsson, for an excellent structured course and Prof. Albania Nissan for helping us with the preparations for the trip. Finally, we would like to express great gratitude towards Sida for the opportunity to establish international collaborations and for making this minor field study possible.

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Nomenclature

BAU	Business as usual
CCGT	Combined Cycle Gas Turbine
CSP	Concentrated Solar Power
EIA	Energy Information Administration
IEA	International Energy Agency
IRENA	International Renewable Energy Agency
OECD	Organisation for Economic Co-operation and Development
OSeMOSYS	Open Source Energy Modelling System
O&M	Operation and Maintenance
PV	Photovoltaic
SDG	Sustainable Development Goal
WtE	Waste to Energy

1.Introduction

Climate change and global warming, arguably humankind's biggest challenges throughout history, originates from excessive use of fossil fuels which emits greenhouse gasses into the atmosphere. To reduce these emissions while meeting the energy demand of today as well as in the future, a transition towards renewable energy sources will be needed (World Commission on Environment and Development, 1987).

Since the vantage point of 1987, with the presentation of the Brundtland report, the United Nations has continually worked towards the goal of a more sustainable world. Following the Brundtland report was Agenda 21, ratified in Rio de Janeiro in 1992 in order to establish a framework for sustainable development (Andrews, 2016, pp. 1-4). One of the most recent developments in this area is the introduction of the SDGs which embodies the core of the Agenda 2030. These goals concretize the needs to be met in order to achieve a world not only characterized by sustainability but defined by it. The new agenda also highlights the need to support vulnerable countries, the majority of which are located in Africa (UN, 2015, pp. 5-7).

Africa is composed of dozens of countries with significant variations regarding economic and social development. One of the main obstacles many share is the need to provide electricity to their population. In sub-Saharan Africa, 62,5 % of the population lacked access to electricity in 2017, thus making lighting and powering up Africa one of the United Nations highest priorities. Egypt on the other hand has an electrification rate of 99,8 %, allowing them to pursue other goals such as SDG 7, providing affordable and clean energy and SDG 13, reducing global climate impact (The Sustainable Development Goals Center for Africa, 2018, p. 2, 71; World Economic Forum, ca. 2017).

According to Egypt's sustainable development strategy, Egypt vision 2030 for energy, the country aims to diversify their energy mix to reach international standards. Furthermore, Egypt regularly works on meeting the growing energy demand from all sectors, while reducing the total amount of emissions emitted from the energy sector (SDSEgypt2030, 2016, pp. 94-96). The country's ambition, according to Egypt's Integrated Sustainable Energy Strategy to 2035, is to reach 20 % renewable energy in their electricity mix by 2022 and 42 % by 2035. As of 2015, the renewable share of the electricity was 8,5 %, where hydroelectric power accounted for 7,5 % (IRENA, 2018a, p. 14, 29).

In line with previously stated ambitions, the government of Egypt is currently building the new city El-Alamein that is called "*the start of a different generation of new cities in Egypt that depends on sustainability*", as stated in the publication from the state information service with the name "*Al - Alamein, from war and destruction to development and construction*" (2018, p. 43).

The publication further describes El Alamein as one of three new cities that the government, together with the private sector, will build as a step to manage the overpopulation that exists in the most densely populated cities like Alexandria and Cairo. The cities will hold residents, industry, universities and agriculture while at the same time focusing on tourism. El Alamein has been named a fourth-generation city because of its focus on sustainability and resource cycles (State Information Service, 2018, pp. 41-48).

In agreement with the approach defined by sustainability and the fact that the city's infrastructure is under construction, the opportunity for investments into utilization of renewable energy sources such as solar, wind, biomass and hydropower are optimal. However, by being the first of its kind in Egypt, there is no information available regarding possible cost-efficient renewable energy solutions that could be applied to El Alamein. The lack of information may influence the decision-makers to create suboptimal solutions for the city's energy system, in terms of minimizing climate impact and creating clean and affordable electricity. More specifically, the lack of available information may lead to wrongful investments that are not cost efficient and sustainable. The information provided by this project could therefore be of great importance in future policy and decision making. Simultaneously this study contributes to cover the gap regarding the overall available information about renewable energy systems in cities within Egypt. These two benefits may together pave the way for the development of future sustainable cities in Egypt, with the city of El Alamein as the focal point.

1.1 Purpose

In line with the United Nations Sustainable Development Goals, the purpose of this field study is to examine the prospects of renewable energy technologies with regards to economic factors in the city of El Alamein.

1.2 Objective

One objective of this project is to identify the electricity demand in the city of El Alamein. In addition, an objective is to identify renewable energy technologies capable of meeting the demand for electricity in the city of El Alamein. Thereafter, an objective is to create an energy system model with the identified technologies that determines the most cost-efficient electricity mix while meeting the demand. In addition, the objective also aims to identify which renewable electricity mix that reduces emissions in the most cost-efficient manner, with reference to a fossil-fuel based equivalent. To conduct a discussion regarding the future use of renewable energy in El Alamein in particular, and Egypt in general, based on the results obtained from this project, will be the last main objective. Furthermore, the project aims to improve the relationship between KTH and Pharos University through collaboration and promote Sustainable Development Goals. There are also goals of a more general character. These include, but are not limited to, the possibility for the project to contribute to the establishment of future policies regarding the energy transition towards renewable energy.

2. Methodology and source discussion

This thesis has mainly been conducted during an eight week stay in Alexandria, Egypt. The work has partly been done in collaboration with Pharos University. A field visit took place during the stay to gather data necessary for the report and to establish an overall understanding of the environment in which our energy system would be implemented.

2.1 Collecting information

To gather information, a combination of two methodologies – literature search and field study – were implemented to provide all necessary information. The literature study was mainly conducted through internet searches to gather information about the technologies, Egypt's electricity production as well as the weather conditions of the examined site. This approach was also used to find information needed for the required assumptions. The main sources of information used, when available, was renowned agencies and organisations such as IRENA, IEA and EIA. When these were insufficient, scientific reports as well as government websites and their documents were largely used.

To complete the gathering of information, a field study involving an interview with construction officials and site examinations were performed. The performed interview was semi-structured, using prepared questions while allowing the interviewees to go into greater detail where they felt appropriate. The interview was conducted through interpreters, and the prepared questions were related to the construction of El Alamein, with focus on the planned capacity, estimated construction completion date and the electricity supply. For the transcript of the interview, see Appendix 1. Moreover, the field visit had the purpose of obtaining a good and realistic picture regarding the circumstances in which the modelled energy system would be implemented. Also, to a further extent, to see if the information that was previously obtained about the site was in line with what was observed when visiting.

2.2 Modelling with OSeMOSYS

The tool used for the modelling of the energy system in El Alamein was OSeMOSYS. This platform was developed through a collaboration between several institutes and organisations such as KTH Royal Institute of Technology and the International Atomic Energy Agency (IAEA). It was first published in 2008 and the application of the modelling tool is stated as *“OSeMOSYS is an open source modelling system for long-rung integrated assessment and energy planning”* (KTH dESA, 2019, pp. 5-8). OSeMOSYS uses linear optimisation to calculate the most cost-efficient energy supply mix, i.e. minimizing the total discounted cost, that meets the defined demand of the examined subject, in this case a city. This is done for every year of the model period and within the operational space defined by the different constraints implemented such as emissions limits or defined renewable share of the energy mix. The modelling system was firstly written in GNU MathProg which is a mathematical language of a high level and uses an approach defined by algebraic expressions and equations (KTH dESA, 2019, p. 8).

OSeMOSYS operates on input data for every parameter, every year of the model period. With this information it optimizes the energy system, as described above, and presents an output consisting of different variables of information regarding the energy system. These so-called variables account for a wide range of factors, such as many different types of costs for different technologies to the rate of production from different technologies each year. This thesis used OSeMOSYS as a tool to produce four scenarios. The scenarios had different constraints such as percentages of renewables for electricity production, or a predetermined production mix. Two scenarios consisted of different percentages of renewables in the energy mix, one represented Egypt's solution for the city's energy system and the last scenario was a reference case that represented business as usual. The scenarios were produced in OSeMOSYS with the purpose of obtaining the variables mentioned earlier and, to a further extent, enable them to be presented with graphs, showing different cost-efficient electricity mixes in the city of El Alamein.

2.3 Analysis

After the completion of the different scenarios, each scenario's electricity mix production and total installed capacity were examined. The behaviour of the different technologies in each mix were thereafter analysed with regards to their constraints, to determine the reason for their contribution. Additionally, the total discounted cost and total amount of emissions were identified for each scenario. These two variables were thereafter compared to the business as usual case for each scenario, and the cost of reducing emissions was identified. This cost was then compared between the scenarios, to determine which electricity mix reduced emissions in the most cost-efficient manner. Lastly, a general discussion was conducted on the different renewable technologies and their potential for future implementation in Egypt.

2.4 Limitations

First and foremost, the project only considered the production of energy in the form of electricity. This implies that energy forms such as heat (and cooling) were not considered directly, but rather seen as being produced by electricity. This means that a certain demand for heat was considered as the equivalent demand for electricity needed for producing that heat. The electricity that was not produced within the city of El Alamein, was considered as if it was produced from a single source and represented the reference case that will be described later in this report. Thirdly, El Alamein was seen as an isolated unit that was not interacting with its surroundings in any matter apart from the incoming flow of electricity.

Furthermore, the project only evaluated and accounted for the development of the energy system in El Alamein from 2020 until 2040, events thereafter were not taken into consideration. It is important to state that the project was performed with an oversimplified approach since the combination of time, resources and the magnitude of the project would not allow a more detailed approach. This influenced the information gathered, especially the information regarding the different technologies.

2.5 Source discussion

As previously mentioned, many of the sources used were recognized agencies and organisations, and the information gathered from these sources was therefore considered to be reliable and of high quality. The information gathered from the other sources, such as research articles, may have inherent deficiencies but despite this possibility, they were also considered reliable since they were, many times, conducted under the supervision of reliable institutes like universities. However, because of the possibility of low-quality information provided from one source, the solution throughout the whole report was to strengthen the information gathered by referring to other sources as well as to ensure the reliability of the information used. The information may sometimes also suffer from being what could be considered outdated but was still used because of the lack of more current information.

The information gathered from the field visit and specifically the interview, has the possibility of suffering from subjectiveness since the person interviewed answered the questions from their own experience and knowledge. Furthermore, the information obtained from the interview went through the process of interpretation, which in itself could consist of problems regarding, for example, subjectiveness and the lack of ability to express the information in English, since the interview was conducted in Arabic. This problem was met by using several interpreters to enable translations from different people and therefore minimizing the risk of misinterpretations. Also, clear and simple questions were used to avoid confusion and vague answers.

3. Background study

This section aims to introduce historical and current information to provide a general understanding of the energy situation in Egypt.

3.1 National energy situation

In 2014/2015 the country's primary energy supply was mainly divided between oil and gas, whom together accounted for 94 %. The remaining percentages were shared among coal, solar, wind, hydropower as well as biofuels and waste, see **Figure 1** (IRENA, 2018a, p. 9).

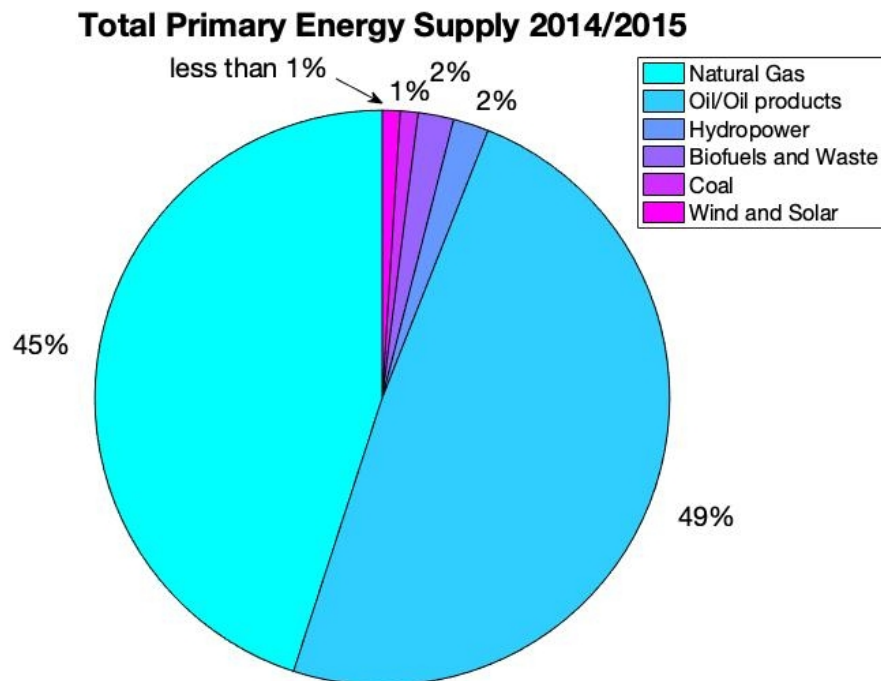


Figure 1. The share between different primary energy sources used in Egypt 2014/2015.
[Data collected from IRENA, 2018a, p. 9].

In the same year of 2015, Egypt's electricity was primarily produced by natural gas, 66 %, and dual-fuel plants that accounted for 19 % of the total production. Dual-fuel plants utilizes two types of fuels when producing electricity, the fuels can vary but usually consists of diesel and natural gas (Hall & Kassa, 2018, p. 7). The rest of the electricity production was divided by hydro and other renewables such as wind and solar as seen in **Figure 2**.

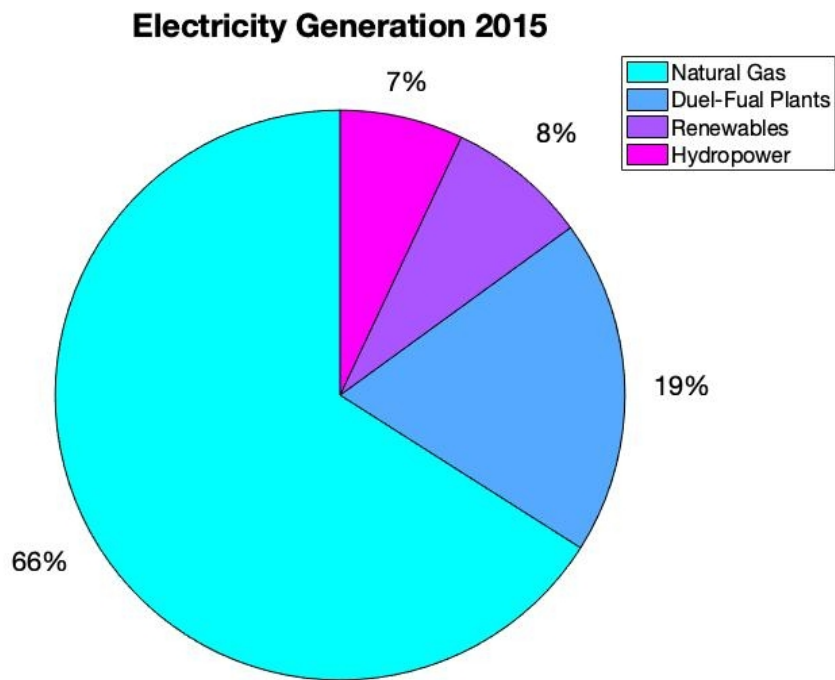


Figure 2. Share of electricity generation from different technologies in Egypt 2015.
 [Data collected from IRENA, 2018a, p. 14].

The total installed capacity for Egypt increased to 38 857 MW under 2015/2016. This capacity was distributed between five different types of sources/plants. These are, presented in descending order of installed capacity, thermal steam plants, CCGT plants, OCGT (open-cycle gas turbine) plants, hydroelectric plants and non-hydro renewable plants (IRENA, 2018a, p. 12).

3.2 Demand distribution

The maximum peak load distribution during 24 hours of the day in 2016/2017 was gathered from the Ministry of Electricity & Renewable Energy in Egypt and is presented in the upper graph in **Figure 3**. Moreover, the lower graph in **Figure 3** is the electricity consumption in Egypt by month in 2016 and was gathered from CEIC: Global Economic Data, Indicators, Charts & Forecasts. The monthly consumption is a representation of the demand distributed each month of the year, where the summer and intermediate months have a larger demand than during the winter for that year.

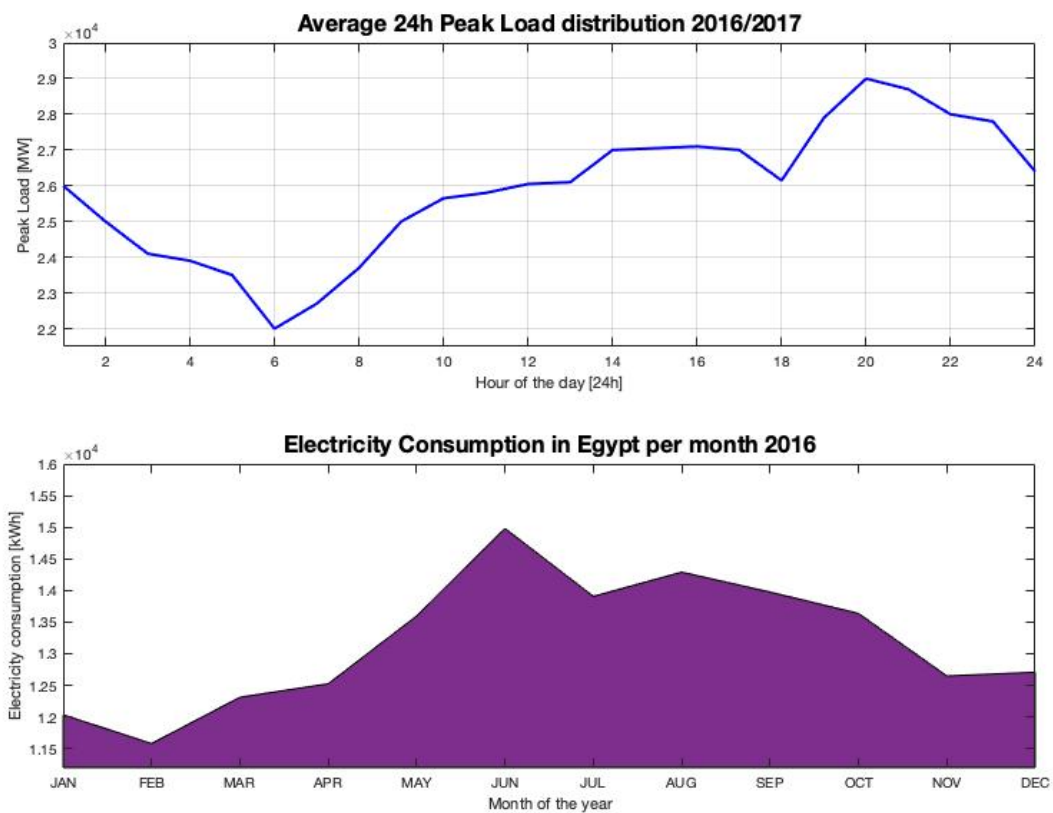


Figure 3. Upper graph: Maximum peak load during 24h in Egypt 2016/2017.
Lower graph: Electricity consumption each month in Egypt 2016.
[Data collected from [CEIC](#) and [EEHC](#)].

3.3 National strategies and goals

The closest and most extensive goal set out by the Egyptian government is the plan of doubling the installed capacity from 50 GW in 2018, to about 100 GW in 2020. Renewable energy will contribute with diversification of this newly planned capacity. The government has committed themselves to implementing another 10 GW of solar and wind by 2022, where renewables would account for 20 % of the electricity mix (IRENA, 2018a, p. 4,7).

ISES to 2035, Egypt's Integrated Sustainable Energy Strategy, contains several aims for the country's future energy development. Among them are aims of meeting the growing demand of energy while decreasing the amount of emissions and pollutants from the sector and diversifying the energy mix. A goal of increasing the production amount from renewable energy from 8,5 % in 2015 to 42 % by 2035, serve as a strategy to fulfil the aims mentioned above. The strategy towards more renewables according to Egypt's Vision 2030 has a target where renewables will account for 35 % of the electricity production by 2030, where wind and solar will account for most of the renewables, see **Figure 4** (IRENA; 2018, p. 6; SDSEgypt2030, 2016, pp. 94-97).

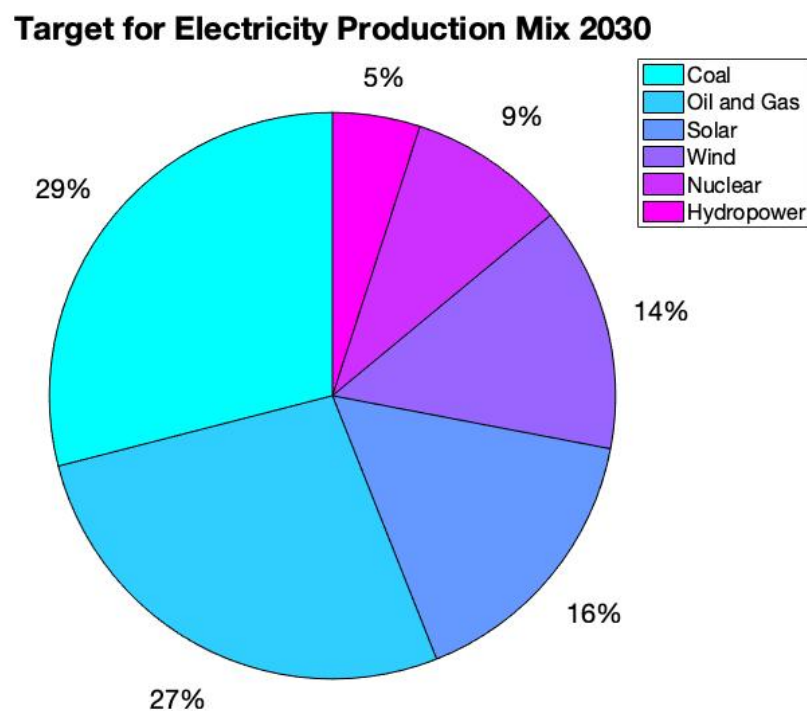


Figure 4. Share of different primary energy sources targeted for electricity production in Egypt by 2030.
[Data collected from SDSEgypt2030, 2016, p. 97].

4. Field study in El Alamein

On the 23rd of March 2019, a field visit to the construction site for the city of El Alamein was made. During the visit one interviews was performed and several locations throughout the site were visited and photographed. The interview served the purpose of providing information for the project and to contribute to an overall understanding regarding the general thoughts and plans about the construction of the city, as well as of the ambitions for the city. The transportation around the construction site and the locations visited, confirmed some of the information gathered from the interview and served as basis for the construction of our limitations and assumptions throughout the project.

During the field visit several observations were made. Firstly, many of the buildings that were close to be finished appeared to be of same height which would therefore implicate that shadows from one building would not necessarily cover the roofs of other buildings. The beaches were not constructed upon and the water close by was not used for any other purpose than for recreation. Thirdly, no indications of renewable energy technologies such as wind turbines could be seen. Lastly the overall construction site seemed to progress at a rapid speed.



Figure 5. Photos from the field visit to El Alamein, showing the ongoing development in the area as well as a photo from one of the interviews being conducted. Photos taken by: Amr Mamdouh, Marko Miletic, Simon Färegård & Erik von Schultz.

5. Local prerequisites

This section contains information specific for the location and future development of El Alamein.

5.1 Population and electricity demand

During an interview conducted on 23rd of March 2019 in El Alamein, the electrical manager Eng. Ahmed Hassan and Eng. Abd El Hamid working on the project El Alamein, stated that the construction of the city is divided into four different phases. The initial phase began in March 2018 and was estimated to be finished in June 2020. When the other three phases will be finished was not known at the time. Each phase was expected to house one million residents, resulting in a total of four million citizens when the city is completed. The available information they had regarding the electricity demand for the different phases were for the first and second phase, which was predicted to be 750 and 1325 MW respectively. As for the total area of the city, it was said to be 70 000 feddan, which equals 294 square kilometres. Moreover, it was stated in the interview that no power stations such as offshore wind power or wave energy will be utilized close to the beaches, since these will be preserved for tourism, see Appendix 1.

5.2 Agriculture and waste management

As stated by Eng. Ahmed Hassan in the interview, there are currently no plans for any type of agriculture in the surrounding area of the city, resulting in no available waste or crop residue generated from agriculture in El Alamein. Furthermore, a recycling plant will be in place once the city is complete, but Eng. Ahmed Hassan did not know the specified capacity or which material the plant would recycle, see Appendix 1. No information regarding the waste management in El Alamein was obtained. However, a waste management study in Egypt mentioned that waste generation in the nearby city of Alexandria was 0,8 kg/capita per day in 1999 (Egyptian Environmental Affairs Agency, 1999, p. 3).

5.3 Wind and solar data in El Alamein

The wind speed data presented in **Figure 6** was gathered from El Alamein in 2017. The wind speed data gathered from <http://www.renewables.ninja/>, which was obtained from NASA, suggested that the monthly average wind speed does not drop below 6,6 m/s. This could be seen as an indicator to why wind energy should be prospected in El Alamein. Furthermore, as seen in **Figure 6**, the most optimal wind harvesting hours occurred during the winter months and until early spring, where wind speeds reached up to 14 m/s. The solar radiation data, also gathered from Renewables Ninja presented in **Figure 7** was gathered from El Alamein in 2017. As seen in the upper graph in **Figure 7**, the solar radiation was more consistent midway through the year and reached peak values of up to 0,3346 kW/m². This could be seen as an indicator to why solar energy should be prospected in El Alamein. Furthermore, as seen in the lower graph in **Figure 7**, the most optimal period to harvest solar energy was throughout summer and in the beginning of autumn. Most of the peak values occurred within this time period. On the contrary, winter period carried fewer sun hours i.e. lower peak values and had more fluctuations that year.

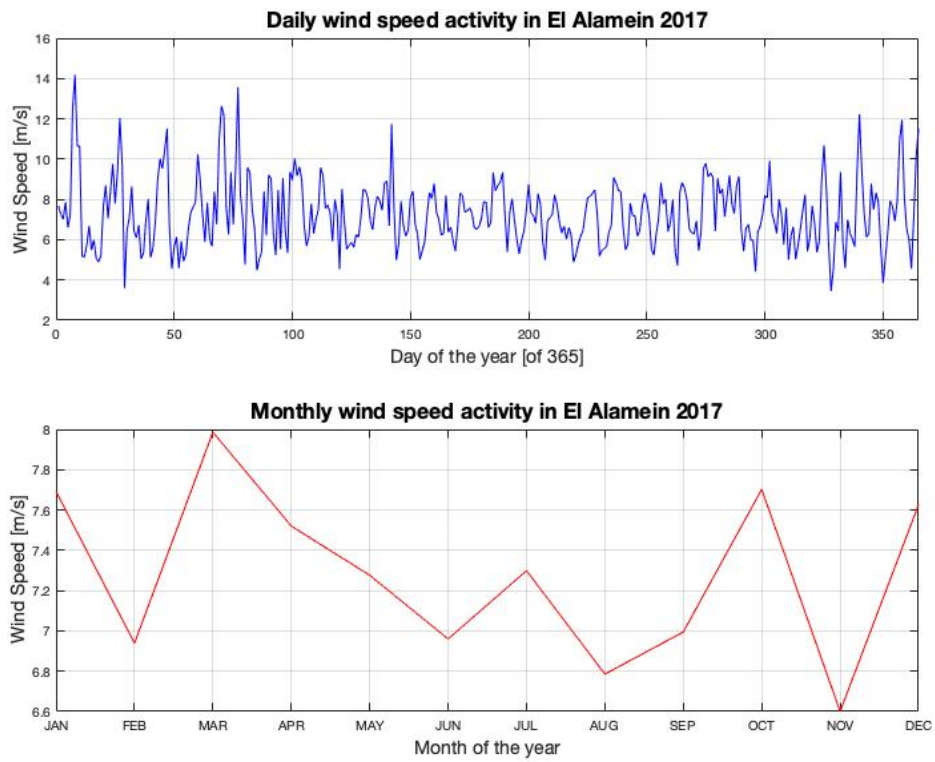


Figure 6. Upper graph: Daily wind speed activity in [m/s] each day of the year.
Lower graph: Monthly wind speed activity [m/s] each month of the year.
[Data collected from <https://www.renewables.ninja/>].

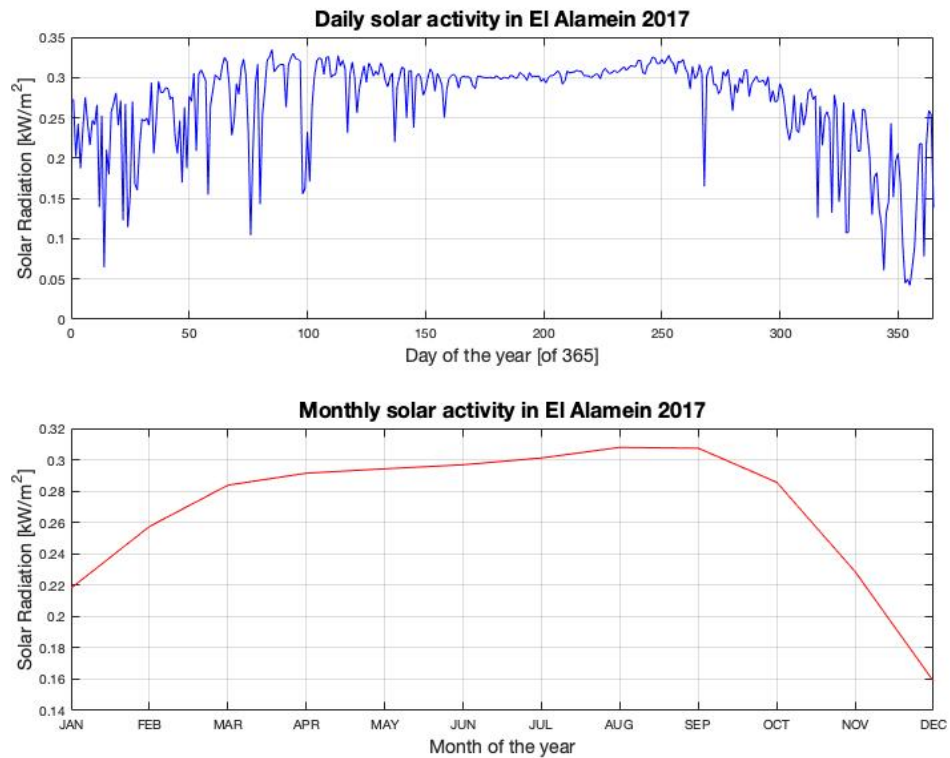


Figure 7. Upper graph: Daily solar activity in [kW/m²] each day of the year.
Lower graph: Monthly solar activity [kW/m²] each month of the year.
[Data collected from <https://www.renewables.ninja/>].

6. Technologies

This section contains information and calculations for the different technologies. The cost of each technology was divided into three separate costs and applied to 2020 as a starting point. The first was the overnight capital cost per installed MW, which consists of a single payment required to launch a project, covering the cost of construction for example. Two additional are the fixed and variable O&M costs. The fixed O&M costs in \$/MW, are the expenses needed for staffing and administrative expenses among others. The variable O&M costs are the expense needed for the production processes, including fuel (EIA, 2016, pp 37-41). The fixed costs for all technologies, despite CSP, were obtained from the same source, as to not let factors like different salaries from different countries affect the outcome when possible. When available, global trends in cost variation over a period of time was used to estimate the future price development except for the fixed cost, which was assumed to be constant.

The information regarding the technologies also contained operational life and capacity factor. The capacity factor is the ratio between the actual production and the productions maximum potential, which was either identified or calculated (University of Massachusetts, 2008). The capacity factors that fluctuated with time were divided into different time periods called time slices, which represented their rate of production during that period. The whole year consists of 24 different time slices, where each time slice represents day or night for a whole month, defining day as the time between 08:00-20:00 and night as the remaining hours. Information regarding emissions will only be provided for the non-renewable technologies, since the emissions from the renewables was estimated to be zero. The technologies listed below will be electricity producing technologies in OSeMOSYS and the reference energy system seen in **Figure 8** was the basis from which the scenarios were applied on.

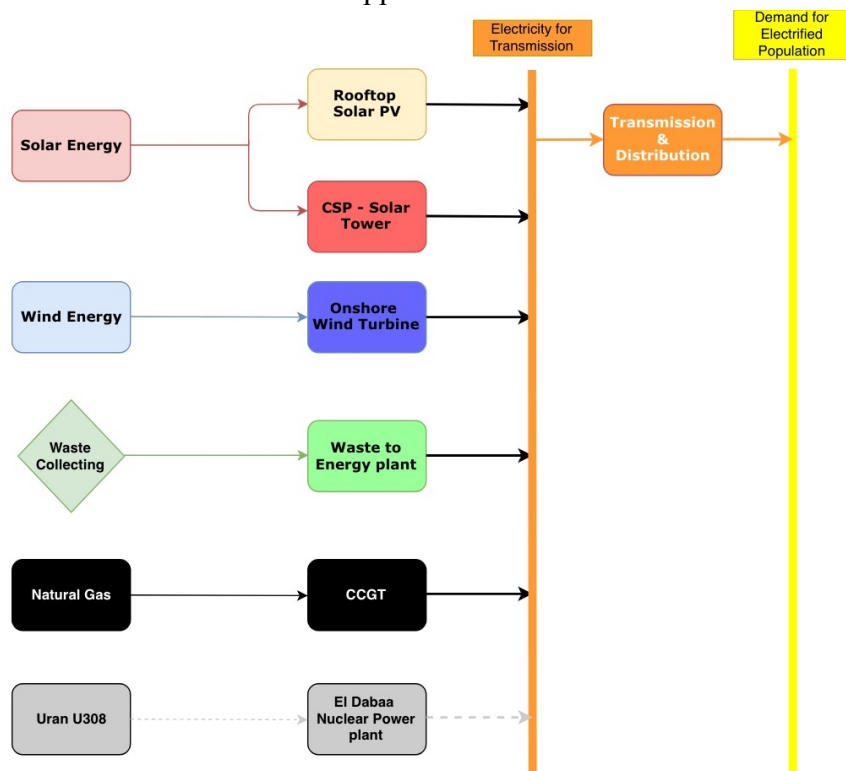


Figure 8. Reference Energy System of technologies and primary energy sources identified for El Alamein.

6.1 Onshore Wind Power



Figure 9. Wind turbines of model Nordex N43 600 aligned in Zaafarana. Photo: Marko Miletic.

There are different types of wind turbines available, the most common one people think of is the horizontal-axis turbine, see **Figure 9**. The mechanism that follows is that the wind forces the blades to rotate, transferring mechanical power on to a rotating shaft, then to the gearbox and lastly to the generator which generates electricity (Heinzen & Reinemann, 2014, p. 1).

Wind power's capital cost was selected to \$1,2 million/MW. This value lies between the estimated global average cost of \$1,5 million/MW and costs in emerging economies, namely India and China are about \$1,0 million/MW. Therefore, a value in between was considered reasonable. The fixed O&M cost was set to \$39 700/MW-year, in line with the information from the US Energy Information Administration, for an onshore wind technology (2016, p. 41). The variable O&M cost for onshore wind power production was set to

\$20/MWh, chosen from the lower spectrum of an average from OECD countries. Moreover, the lower value of the spectrum was chosen since the average in other regions, including both India and China, were below the values of the chosen spectrum (IRENA, 2018b, pp. 93-94, 108). The operational life time for the wind power plants was set to 20 years, equal to the life time of Nordex N43 600, a wind turbine model used in Egypt (Rehman et al., 2003, p. 578; TheWindPower, 2018).

Both capital and variable cost will change in line with the estimated 70 % average decline in installed cost of wind farms from IRENA's report on renewable power generation costs in 2017, measured between 1983 to 2017 (2018b, p.94). Assuming a constant yearly decline, the percentage was calculated to 3,48 % by using the following equation

$$\text{Annual \% change} = \left(\frac{\text{Final cost}}{\text{Initial cost}}^{\frac{1}{\text{Years Difference}}} - 1 \right) * 100, \quad (1)$$

the different capital costs and variable O&M costs throughout all years are presented in Appendix 2.1 and Appendix 2.2. Onshore wind turbine's capacity factor was estimated using the model from Renewables Ninja. The inputs that were used to generate these capacity factors were the location of El Alamein, the year 2017 which was assumed to reflect all years, the hub height of 80 meters (that is the height where the blades are centred) and the turbine model Nordex N43 600. The model used the NASA dataset of MERRA-2 (Renewables Ninja, 2018?). Moreover, the capacity factor for the day and night was assumed to be equal each month. Lastly, the capacity factor obtained from the year 2017 was adjusted for each time slice to account for historical values, according to

$$Adjusted\ CF_{TimeSlice} = \overline{CF}_{Historical} * \frac{\overline{CF}_{TimeSlice,RN}}{\overline{CF}_{yearly,RN}}, \quad (2)$$

where the $CF_{Historical}$ was calculated as the mean value of the yearly average capacity factor from 2000 to 2016, resulting in $CF_{Historical} = 0,2867$. The mean value of $CF_{yearly,RN}$ in 2017, obtained from Renewables Ninja was $CF_{yearly,RN} = 0,267$. With the $CF_{Historical}$ calculated and the $CF_{TimeSlice,RN}$ identified from Renewable Ninja as the capacity factor each month, using equation (2) the values of the adjusted capacity factors for each time slice were obtained, see Appendix 2.3. Furthermore, an area of 656 km² along the Suez Canal has been devoted to implementing a wind farm with a capacity of 3000 MW, which equals an area usage of 0,2186 km²/MW (Wind Energy the Facts, cs. 2007). All initial parameters for 2020 that are presented in the paragraphs above are gathered in **Table 1**.

Table 1. Input parameters for onshore wind turbine in the initial year of 2020 in OSeMOSYS

Input parameters for 2020	Onshore Wind Turbine
Capital Cost (\$/MW)	1 200 000
Fixed O&M Cost (\$/MW-year)	39 700
Variable O&M Cost (\$/MWh)	20
Operational Life (years)	20
Average Capacity Factor (%)	26,7

6.2 Solar PV

Photovoltaics (PV) generate electricity by utilizing a combination of photoelectric materials and light. The light causes the material to absorb photons, leading to a release of electrons which is used as electricity. Solar PV can be installed in both large and small scales, resulting in it being applicable on large PV farms and small mini-grids (IRENA, 2018c; Knier, 2008).

The capital cost for solar PV was set to \$1,5 million/MW, which is in the lower spectrum of the cost range according to IRENA's report Renewable Power generation Cost in 2017 (2018b, p. 66), where the total installed cost in 2017 for different countries range between \$1 million/MW to \$4,5 million/MW. The chosen fixed O&M cost was \$23 400/MW-year, taken from a fixed PV model, and the variable O&M cost was set to \$0/MWh in accordance with the same model (EIA, 2016, p. 11). The technology's operation lifetime was set to 30 years (IRENA, 2018c). The cost trend for the solar PV capital cost was assumed to continue in line with the 32 % decrease between 2013 and 2017 in South Africa, resulting in an annual decrease of 9,19 %, see equation (1) (IRENA, 2018b, p. 72). To see the capital cost for each year between 2020-2040, see Appendix 3.1.



Figure 10. Pharos university's solar PV on the faculty of engineering's rooftop. Photo: Simon Färegård.

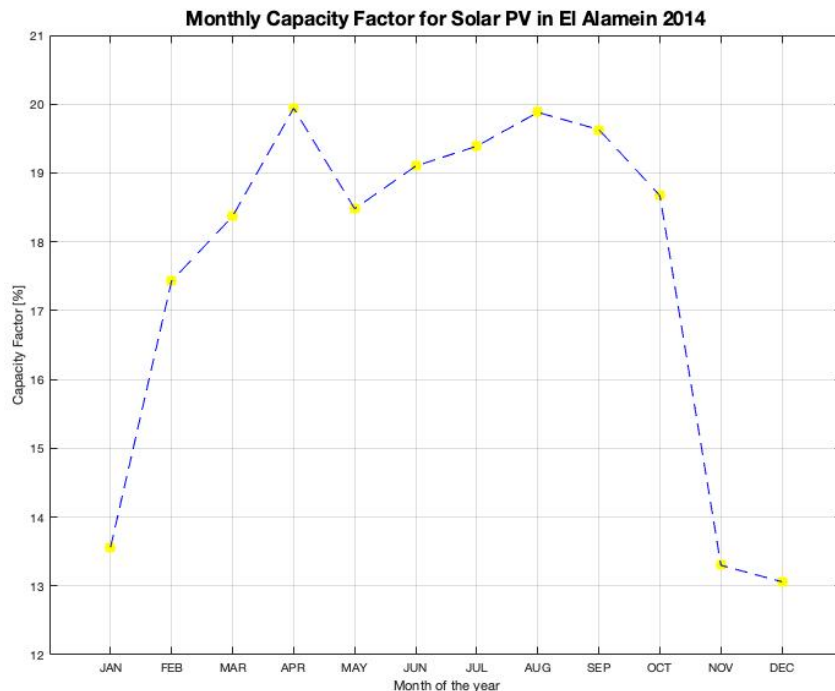


Figure 11. Capacity factor for solar PV each month of the year in El Alamein 2014. [Data collected from <https://www.renewables.ninja/>].

The average capacity factor was set to 17,6 %, i.e. the global average for utility scale PV systems in 2017 (IRENA, 2018b, p. 66). By taking the change in electricity production between days and nights, as well as the change in capacity factor between seasons into account, the capacity factor for each time slice was identified. The monthly change in capacity factor was obtained from Renewables Ninja with the default inputs for solar PV in El Alamein 2014, by calculating the difference in percentage between the yearly mean value and the value each month. This percentage increase or decrease from the average value was then applied to the previously mentioned average capacity factor, 17,6 %, resulting in a more realistic average capacity factor for each month, see **Figure 11**.

To distribute the monthly capacity factor between day and night, the production for each time slice during 2014 from El Alamein was summarized using Solar PV data from Renewables Ninja. To get the fraction of production during night or day, the production during these periods were summarized and divided by the months total production. This value was then multiplied by two, to account for the time difference between the two production periods. Finally, the value was multiplied by the monthly average capacity factor as presented in the following equation

$$CF_{TimeSlice} = ProductionFraction_{day/night} * CF_{monthly,avg} * 2. \quad (3)$$

The capacity factors for each time slice is presented in Appendix 3.2. Finally, the area utilized by solar PV was based on Fraunhofer's predication for Germany. The prediction stated that 200 GW of solar PV's will require an area of 1000 km², with an efficiency of 19 % assumed for the panels. This results in the installed capacity per area being 0,2 kW/m² (Wirth, 2019, p. 39). All initial parameters for 2020 that are presented in the paragraphs above are gathered in **Table 2**.

Table 2. Input parameters for solar PV in the initial year of 2020 in OSeMOSYS

Input parameters for 2020	Solar PV
Capital Cost (\$/MW)	1 500 000
Fixed O&M Cost (\$/MW-year)	23 400
Variable O&M Cost (\$/MWh)	0
Operational Life (years)	30
Average Capacity Factor (%)	17,6

6.3 Waste to Energy plant



Figure 12. Photo of a Waste to Energy plant. [Source: ÅF].

There are different methods used when harnessing energy from biowaste, this project focused on the method of burning waste to generate electricity. The biowaste is burned to produce heat which evaporates water in a boiler to produce steam. The steam then travels through a steam turbine which transfers mechanical energy to a generator who then generates electricity (EIA, 2018). The emissions from burning biowaste mainly consist of carbon dioxide but the

biowaste is seen as a renewable energy source when burned, with net zero emissions since the CO₂ emitted is the same amount captured by the bio-waste, i.e. no new emissions are introduced to the atmosphere. The other emissions for the combustion such as SO_x are assumed to be removed by scrubbers, a flue gas cleaning method (Udomstri, 2011, p.12). Since there were no plans for agriculture in El Alamein, the biowaste needed for the process only derived from municipal solid waste.

A majority of the parameters needed for the OSeMOSYS model for the waste to energy plant were taken from an assessment regarding the prospects of using biomass to produce electricity published by the United States Department of Agriculture. The capital cost was \$3,860 million/MW and the variable O&M cost, assuming that the waste provided to the plant was supplied for free, was \$5/MWh. The fixed O&M cost was \$110 000/MW-year and the operational lifetime was set to 20 years (Nderitu, et al., 2014). No major trend regarding a change in price for electricity produced by biomass was identified between 2010 and 2017 by IRENA (2018b, pp. 34-35). Therefore, the capital cost and variable O&M cost was selected to remain the same from 2020 to 2040. Using the same capacity factor for the biomass plant as for a coal power plant resulted in it being 85 % (EIA, 2016, p. 41; Nderitu, et al., 2014). The efficiency of electricity production from the incineration process was set to 25 %, since only the electricity production was taken into account (Batool, et al., 2016). All initial parameters for 2020 that are presented in the paragraphs above are gathered in **Table 3**.

Table 3. Input parameters for waste to energy plant in the initial year of 2020 in OSeMOSYS

Input parameters for 2020	Waste to Energy plant
Capital Cost (\$/MW)	3 860 000
Fixed O&M Cost (\$/MW-year)	110 000
Variable O&M Cost (\$/MWh)	5
Operational Life (years)	20
Average Capacity Factor (%)	85

6.4 Concentrated Solar Power – Solar Tower



Figure 13. Pharos University's prototype of a CSP – solar tower built on the faculty of engineering's rooftop.
Photo: Erik von Schultz

There are many different methods to harvest solar energy through concentrating solar power (CSP); the main method considered in this study was solar tower system. The system uses sun tracking mirrors which reflect solar light onto a central receiver that contains a heat transfer fluid. The fluid is then heated by the concentrated solar power from the mirrors and is used to produce steam which will pass through a steam turbine and lastly a generator, where electricity is generated (SolarPACES, 2018).

For this project, a CSP plant with a solar tower with 9 hours of storage that uses molten salt for the thermal storage was considered. The capital cost was set to \$7,88 million/MW based on the cost from 2012. The variable O&M cost was set to be \$31/MWh, which reflects the cost for 100 MW solar tower with 9 hours of storage in 2012 and the fixed O&M cost for a solar tower was set to \$65 000/MW-year (IRENA, 2013a, pp. 61-63). The operational lifetime was set to 25 years (IRENA, 2013b, p. 63). Furthermore, the costs presented above were in line with those presented by IRENA in “Renewable power generation cost 2017” (2017, pp. 77-86). All these values are not specific for Egypt, but since the technology of CSP is yet in an early stage of development, these were chosen as the most reliable and usable for Egypt.

As for the cost trend of CSP, it was assumed to be in line with the one presented by IRENA in “The power to change: Solar and Wind Cost Reductions Potentials to 2025” where they assume an overall cost reduction of 43 % for solar towers between the years 2015 to 2025, equal to a yearly percentage decline of 5,46 %, see equation (1) (2016, p. 99). This reduction of cost was applied to the variable O&M cost as well as the capital cost, but not the fixed cost since it was assumed to be constant as for the other technologies concerned in this report. See Appendix 4.1 for the capital costs each year and Appendix 4.2 for the variable O&M costs each year.

The average capacity factor for the specific CSP Solar tower was chosen to 54 %, but varies in reality between 50-60 % (IRENA, 2012, p. 5,20). To mimic reality, the average capacity factor each month has been set to follow the trend of monthly solar activity seen in **Figure 6** and is presented in Appendix 4.3. Furthermore, since CSP is a solar technology, the capacity factors will differ from day and night since the available sun hours are different. From a CSP production in Spain, the average activity ratio during 24 hours of production was 67,35 % during the day hours (8:00-20:00) and 32,65 % during the night hours (20:00-08:00) (Capellán-Pérez & de Castro, 2018, p. 7).

As the time slices are divided into months and as well as day and night, the final step was to calculate the specific capacity factor for days and nights each month using equation (3) where the *ProductionFraction* is 0,6735 when calculating the capacity factor during the day and 0,3265 during the night. All calculated capacity factors are presented in Appendix 4.3 and all initial parameters for 2020 that are presented in the paragraphs above are gathered in **Table 4**.

Table 4. *Input parameters for concentrated solar power – solar tower in the initial year of 2020 in OSeMOSYS*

Input parameters for 2020	Concentrated solar power (CSP)
Capital Cost (\$/MW)	7 880 000
Fixed O&M Cost (\$/MW-year)	65 000
Variable O&M Cost (\$/MWh)	31
Operational Life (years)	25
Average Capacity Factor (%)	54

6.5 Nuclear Power plant

A nuclear power plant operates by letting neutrons split the atoms in uranium into two smaller atoms. The process is called fission and releases large amounts of heat at the same time. The heat is then utilized by heating liquid water into hot steam, which then passes a steam turbine and generator that generates electricity. The power plant relies heavily on water and uranium to operate (Ontario Power Generation, n.d., pp.2-3).



Figure 14. Photo of a nuclear power plant. [Source: Pixabay].

In February 2015, Russian President Vladimir Putin and his Egyptian counterpart, President Abdel-Fattah al-Sisi, agreed upon the joint operation of building Egypt's first nuclear power plant in the outskirts of El Alamein. The Russian nuclear company ROSATOM is in charge of the construction which began in 2017. They stated that the plant will consist of four reactors, each with a capacity of 1200 MW (4800 MW in total). The first reactor is expected to operate in 2026 and will provide the national grid with 1200 MW and the remaining three reactors will be online in 2028 (Shay, 2018, pp.4-7). This was confirmed during an interview conducted 23rd of March 2019, where electrical manager of El Alamein, Eng. Ahmed Hassan indicated that the first phase of El Dabaa nuclear plant will begin commercial operation in 2026, and that the grid from El Dabaa to El Alamein already was constructed and ready to operate, see Appendix 1.

The El Dabaa nuclear project will receive investments from the Russian government and the private sector. Russia will loan Egypt \$25 billion, which accounts for 85 % of the total capital cost, with the remaining funds provided from the private sector. This results in a total installment cost of roughly \$29,41 billion, equivalent to a specific capacity cost of about \$6,13 million/MW. Furthermore, the fixed O&M cost are estimated to be \$100 280/MW-year, according to the US Energy Information Administration, which was within the scope of similar capacity sized nuclear power plants (EIA, 2016, p. 41; INGÉROP, 2013, p. 138). The variable O&M costs were gathered from a South African nuclear cost study, which refers to a cost analysis made by Moscow Bank of the Russian nuclear industry. The study presents fuel cost at 0,80 US cents per kWh and operation and engineering maintenance cost at 1,93 US cents per kWh. After conversion, this results in an overall total variable O&M cost of \$27,3/MWh (INGÉROP, 2013, p. 59). Furthermore, ROSATOM has announced that service for the reactors will be provided for 60 years which was assumed as the plant's operational life (Shay, 2018, p. 1). Lastly, the capacity factor of a ROSATOM branded nuclear plant was determined to 83 % in 2017 (ROSATOM, 2018).

To analyse the trend of the variable O&M cost (including fuel-cost) over the last 10 years, historical costs of uranium U308 has been used to predict the cost fluctuation at a steady percentage rate per year for the variable O&M costs from 2026 until 2040. From **Figure 15**, it can be observed that the cost of \$45/lb of uranium U308 in 2010 has decreased to \$29/lb as of 2019. This equals a total cost reduction of roughly 36 % over a ten-year period, or an annual cost reduction of 4,36 % according to equation (1).

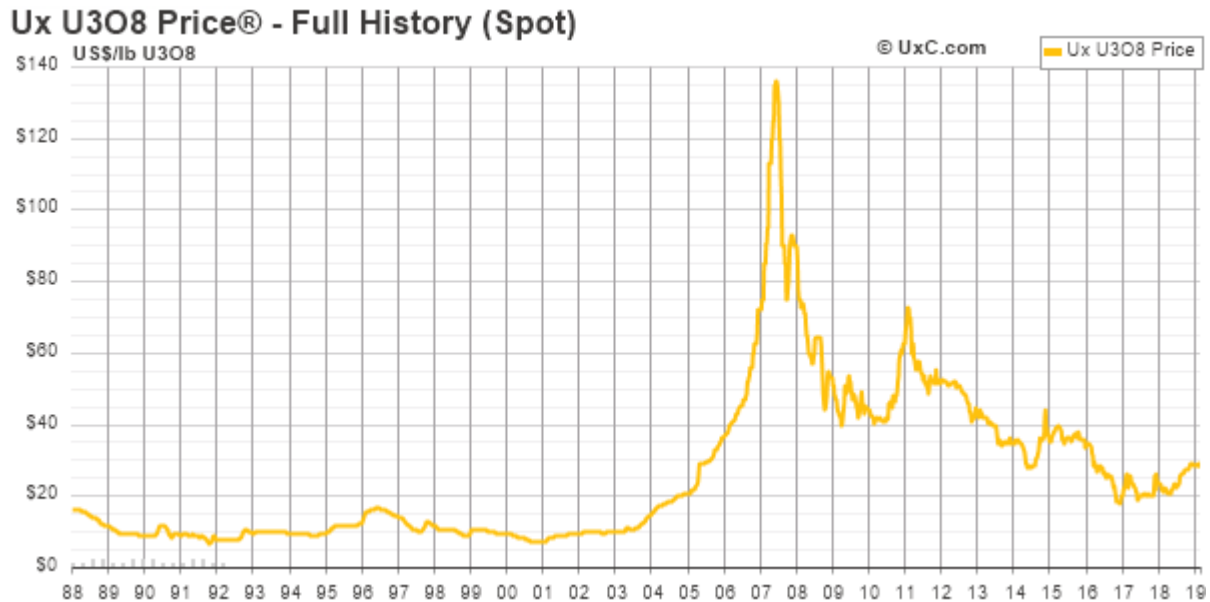


Figure 15. Price history of uranium U308 from 1988 until 2019. [Source: UxC, LLC <https://www.uxc.com/>].

This percentage reduction will be used when predicting future variable O&M cost trends for El Dabaa nuclear power plant from 2026 to the year of 2040 and was included in the calculations that are found in Appendix 5.1. The capital investment cost will not be affected by this trend since the cost of the whole operation was determined beforehand and was seen as the final price throughout all the years. The nuclear power plant will not emit any CO₂-equivalents while producing electricity. Moreover, the chosen variable O&M cost was an estimation, since the project was a first of its kind in Egypt and was reasonably set within the scope of cost suggestion from other similar projects (Rothwell, 2016, p. 16). All initial parameters for 2026 that are presented in the paragraphs above are gathered in **Table 5**.

Table 5. Input parameters for nuclear power plant in the initial year of 2020 in OSeMOSYS

Input parameters for 2026	El Dabaa nuclear power plant
Capital Cost (\$/MW)	6 127 451
Fixed O&M Cost (\$/MW-year)	100 280
Variable O&M Cost (\$/MWh)	27,3
Operational Life (years)	60
Average Capacity Factor (%)	83,29

6.6 Natural Gas CCGT (BAU)



Figure 16. Photo of a natural gas plant. [Source: Unsplash].

A combined cycle gas turbine produces electricity with two turbines, driven from gas and steam respectively. Compressed air and gas are ignited, creating a hot mixture that rotates the gas turbine which converts the mechanical energy to electrical energy through a generator. Excess heat from this process is captured and used to create steam that spins a steam turbine, yet again converting mechanical energy to electrical energy (Ramireddy, 2012).

The capital cost for a CCGT plant in Egypt ranged between about \$0,6 - \$1,2 million/MW, while most capital costs seemed to be around \$0,9 million/MW. The fuel cost, i.e. cost for natural gas, was expressed as an interval and ranged between \$10,2/MWh to \$27,3/MWh in 2016 (Abokersh, et al., 2016, s. 13). In west Africa, the capital cost was about \$1,0 million/MW and the variable O&M cost, excluding the fuel cost, was \$2,9/MWh (IRENA, 2013b, s. 26, 63).

In line with these numbers, capital cost was selected to \$1,0 million/MW and variable O&M cost, excluding fuel cost, was selected to \$2,9/MWh. The fuel cost was set to the average in 2016, which was \$18,75/MWh. The fuel cost together with the previous mentioned variable cost gives the total variable O&M cost of \$21,65/MWh. Furthermore, the fixed O&M cost was set to \$11 000/MW-year. One could also see that the variable cost (excluding fuel cost) proposed by EIA was \$3,5/MW which was in line with the number chosen in this report (EIA, 2016, s. 41). Operational life for the CCGT plant was set to 30 years and the capacity factor was assumed to be 85 %, equal to that of the year 2013, throughout all the years (IRENA, 2013b, s. 63).

The fixed cost was assumed to be constant throughout the timespan, as earlier described. The variable cost, excluding the fuel cost, and the capital cost were assumed to be constant for all years as well. However, predictions for 2035 shows a significant increase in price for natural gas which would range between \$24,2/MWh and \$42,7/MWh which equals an average value of \$33,45/MWh. Because of this, the fuel price, and to a further extent the variable O&M cost, were assumed to follow this increase with a constant yearly percentage of 2,9 %, see equation (1), until the year of 2040. See Appendix 6.1 for the total variable O&M cost for each year. All these assumptions are in line with a prediction made by Fraunhofer in collaboration with the German embassy in Egypt in the report “Electricity cost for renewable energy technologies in Egypt” (2016, s. 16, 23).

Lastly, the amount of greenhouse emissions from the CCGT plant, accounting for the whole life cycle, was set to be 584 kg CO₂-eq/MWh. This number was of the same magnitude as those of previous studies (Agrawal, et al., 2013, s. 1157, 1162). All initial parameters for 2026 that are presented in the paragraphs above are gathered in **Table 6**.

***Table 6.** Input parameters for natural gas CCGT plant in the initial year of 2020 in OSeMOSYS*

Input parameters for 2020	Natural gas CCGT
Capital Cost (\$/MW)	1 000 000
Fixed O&M Cost (\$/MW-year)	11 000
Variable O&M Cost (\$/MWh)	2
Operational Life (years)	30
Average Capacity Factor (%)	83,29
Emissions (kg CO ₂ -eq/MWh)	584

7. Preparations for OSeMOSYS

This chapter uses the previously presented information to make necessary calculations for the different inputs needed for OSeMOSYS.

7.1 Specified annual demand profile

The energy demand during the year changes between the different time slices. The specified electricity demand for each time slices was based on the peak load and electricity demand, see **Figure 3**. The specified demand for each time slice represents the fraction of the total demand during that time slice. The monthly fraction of the total demand was calculated by dividing the monthly values, obtained from the lower graph in **Figure 3**, with the total demand of the year. Moreover, the fraction of the demand during the day and night was calculated by dividing the demand during the day and night, obtained from the upper graph in **Figure 3**, with the total demand for one day. The monthly fractions and the day part fractions were then multiplied with each other, resulting in a specified demand for each time slice, see Appendix 7.1.

7.2 Population and electricity demand

As stated earlier in this report, the first phase was said to have a capacity of 750 MW for the one million people living there, and 1325 MW for the two million residents in the second phase. The first assumption for the electricity demand and population was that there is two years between every phase, with the project starting in 2018 and the first phase being done 2020, with an initial population of 500 000 as soon as it is completed. The population will increase linearly to the completion of the second phase where the city's population will be one million. Thereafter, the population increases by one million for each completed phase, reaching four million by 2028. Regarding the population growth beyond 2028 for El Alamein, it was chosen to be 1,99 % per year in line with data from the UN for Alexandria as seen in **Figure 17** (WorldPopulationReview, 2018).

It was also assumed that the used capacity at the completion of the first phase was 375 MW, half of the 750 MW needed for the first million. Thereafter the capacity is growing linearly to 2022 where the capacity is 750 MW for the one million residents. Within this capacity, there is a “start capacity” which is needed for various community services such as street lights. Because of this “start capacity”, it was assumed that the increase between the first and second phase, 575 MW, was more appropriate to use as the estimation for the increase in the capacity needed for each phase and for each inhabitant.

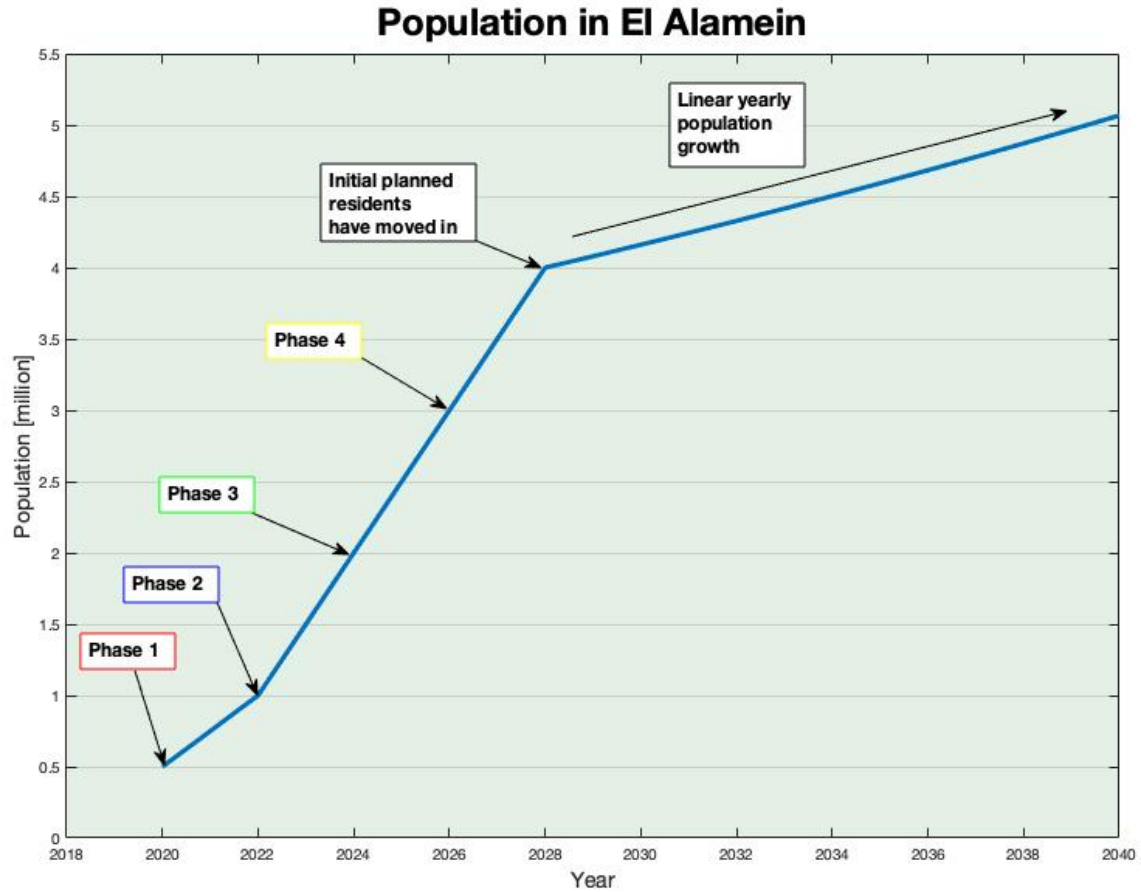


Figure 17. Population in El Alamein each year from 2020 to 2040.

This resulted in the capacity needed being 0,575 kW/capita. It was also assumed that the demand of each phase, and each year, was exactly met when the capacity was producing electricity all year, all the time, resulting in a somewhat high demand. With the above assumptions and with the capacity per capita at 0,575 kW, the demand for each year after 2022 (where the demand is 750 MW) is given by the following equation

$$\text{Demand in MWh per year} = \left(\text{Population per year} * 0,575 \frac{\text{kW}}{\text{capita}} * 8760 \right) / 1000, \quad (4)$$

where 8760 is the total amount of hours in one year. To observe the total electricity demand for each year, see Appendix 7.2.

7.3 Available rooftop area for Solar PV

The solar PV potential was limited by the amount of available area on rooftops within the city as a limitation for the technology in OSeMOSYS. To estimate this area, the method used calculates the rooftop area (rooftop area in square meters per capita) using the population density (population per square kilometre) together with constants derived from a dataset of 1500 samples as shown in the equation

$$\text{Rooftop area per capita} \left[\frac{\text{m}^2}{\text{capita}} \right] = 172,3 * \text{population density}^{-0,352}, \quad (5)$$

which was gathered from the US Energy Information Administration (IEA, 2016, pp. 1-2). Furthermore, the value for population density was needed to calculate the rooftop area per capita and was obtained by dividing the total amount of residents by the total area of El Alamein, as seen in the equation

$$\text{Population density} \left[\frac{\text{residents}}{\text{km}^2} \right] = \frac{\text{Total amount of resident}}{\text{area of the city}}. \quad (6)$$

With values obtained from the interview in El Alamein (see Appendix 1), the population density was calculated to 13 605 residents/km² by using the population in the year of 2028. This value was used in equation (4) to calculate the rooftop area per capita to 6,04 m²/capita. Lastly, the total available rooftop area each year was obtained by multiplying the rooftop area per capita with the population each year shown in **Figure 17** and resulted in 6 040 000 m² available rooftop area for the first year, i.e. 2020. See Appendix 7.3 for the available rooftop area each year.

7.4 Maximum Solar PV utilization

The maximum amount of installed capacity each year for solar PV was calculated with the assumption that all available rooftop area would be used for solar PV and by using an equation as follows

$$\begin{aligned} & \text{Maximum capacity each year [GW]} \\ &= (\text{available rooftop area each year} [\text{m}^2] * \text{capacity per area} [\text{kW}/\text{m}^2]) / 10^6. \end{aligned} \quad (7)$$

Where the *capacity per area* is 0,2 kW/m² and the *available rooftop area each year* can be seen in Appendix 7.3. Lastly, the maximum capacity for solar PV each year can be seen in Appendix 7.4.

7.5 Waste generation for Waste to Energy plant

The energy output for the biomass fuel was selected to 14,31 MJ/kg, which was an average value concluded after comparing four different food groups (Tanai, 2016). It was assumed that all of the waste generated within the city is transported to the recycle plant, where all of the organic matter is separated from the other waste. The separation of the organic matter was implemented to allow the technology to be defined as a renewable. The value of waste generation was chosen to 0,8 kg/capita per day, based on the amount of waste generation in Alexandria presented in the local prerequisites. By multiplying the waste generation with the population each year and amount of days in a year, the total waste generated each year was calculated through

$$\text{Total Waste Generated [kg]} = 0,8 \frac{\text{kg}}{\text{capita,day}} * \text{population} * 365 \text{ days.} \quad (8)$$

Organic matter accounts for 56 % of all municipal solid waste generated each year in Egypt (Elfeki & Tkadlec, 2014, p. 4). To calculate the total organic waste generated, the waste generated each year in equation (8) was multiplied by a factor of 0,56 and is presented in Appendix 7.5. To calculate the potential heat value available for electricity production each year, the generated organic waste was multiplied with the average heat value for organic waste at 14,31MJ/kg and the efficiency of the power plant at 25 % as shown below

$$\begin{aligned} & \text{Electricity production from organic waste each year [PJ]} \\ & = \text{Organic waste generated each year [Mton]} * 14,31 \left[\frac{\text{MJ}}{\text{kg}} \right] * 0,25 . \end{aligned} \quad (9)$$

The results obtained from equation (9) are presented in Appendix 7.6 and represent the maximum possible amount of electricity generated from the waste to energy plant each year for the model.

7.6 Other input parameters

The used modelling tool, OSeMOSYS, required input parameters regarding transmission and distribution losses and operational lifetime, discount rate as well as for the depreciation method. The transmission and distribution losses were set to 11 % which has been the value for the last three years (World Bank, 2018). Also, this was set to be constant since there was no definite trend to be seen in Egypt since 1971. The discount rate was set to the default value in OSeMOSYS, which is 5 %. Lastly the depreciation was assumed to be linear for the technologies.

8. Scenarios

This part is dedicated to defining the different scenarios for El Alamein's energy system. Each scenario is a vision for a future energy system for El Alamein, deciding the share of electricity production between the different technologies. A business as usual scenario was constructed to act as a common point for comparison of costs and emissions. One scenario was constructed to represent Egypt's solution for El Alamein's energy system, whereas the two remaining scenarios compare different renewable energy technologies while meeting a specified minimum production target each year.

8.1 Reference case - Natural Gas (Business as usual)

The reference scenario was based on Egypt's current electricity mix, where a majority of the electricity production comes from natural gas, see **Figure 2**. Therefore, when comparing to this scenario, the comparison in costs and emissions will be done entirely against a CCGT plant fuelled by natural gas.

8.2 Scenario 1 - Nuclear and Natural Gas (Egypt's solution)

The first scenario will begin its operations with the previously mentioned CCGT i.e. natural gas in 2020 and then later introduce El Dabaa nuclear power plant where the first reactor comes online 2026. As mentioned in the interview with Eng. Ahmed Hassan, El Alamein aims to utilize the nuclear plant when it comes online (see Appendix 1). For this reason, a cost and emission comparison will be done against a combination of CCGT and nuclear power plant.

8.3 Scenario 2 – 50 % Renewables

The second scenario, starting in 2020, was determined to provide 50 % of the electricity in El Alamein in 2040 from renewable energy sources, inspired by Egypt's strategic aims and goals. These aims and goals, as shown in **Figure 18**, will be the renewable share required to be met and which increases linearly in between each goal.

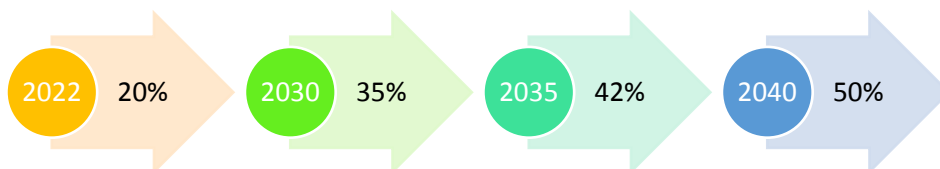


Figure 18. Strategic renewable share targets to achieve certain years of the time period in scenario 2.

8.4 Scenario 3 – 100 % Renewables

The third scenario for the OSeMOSYS model will be more ambitious. It follows the same structure as scenario 2 with the difference that the goal for the electricity production mix in 2040 was set to 100 % renewables. The increase between year 2020 to 2022 will be linear to meet the first sub goal, see **Figure 19**. After 2022, where it will be 20 % renewables, the increase will be linear to the year 2040 where it will be 100 %. This will insure that all sub goals will be met.

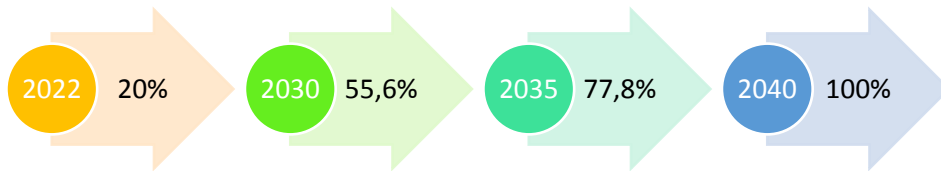


Figure 19. Strategic renewable share targets to achieve certain years of the time period in scenario 3.

9. Results

In this part the relevant variables gathered from OSeMOSYS will be presented in graphs and tables for each scenario, followed by a short description.

9.1 Reference case – Natural Gas (Business as usual)

The production mix of the reference scenario consisted solely of combined cycle gas turbine using natural gas, as seen in **Figure 20**. It can also be seen that the total production increased every year in line with the rising population, see **Figure 17**. The demand increased quickly until 2028, afterwards it can be seen that the demand continues to grow all the way to 2040, but not as rapidly as before.

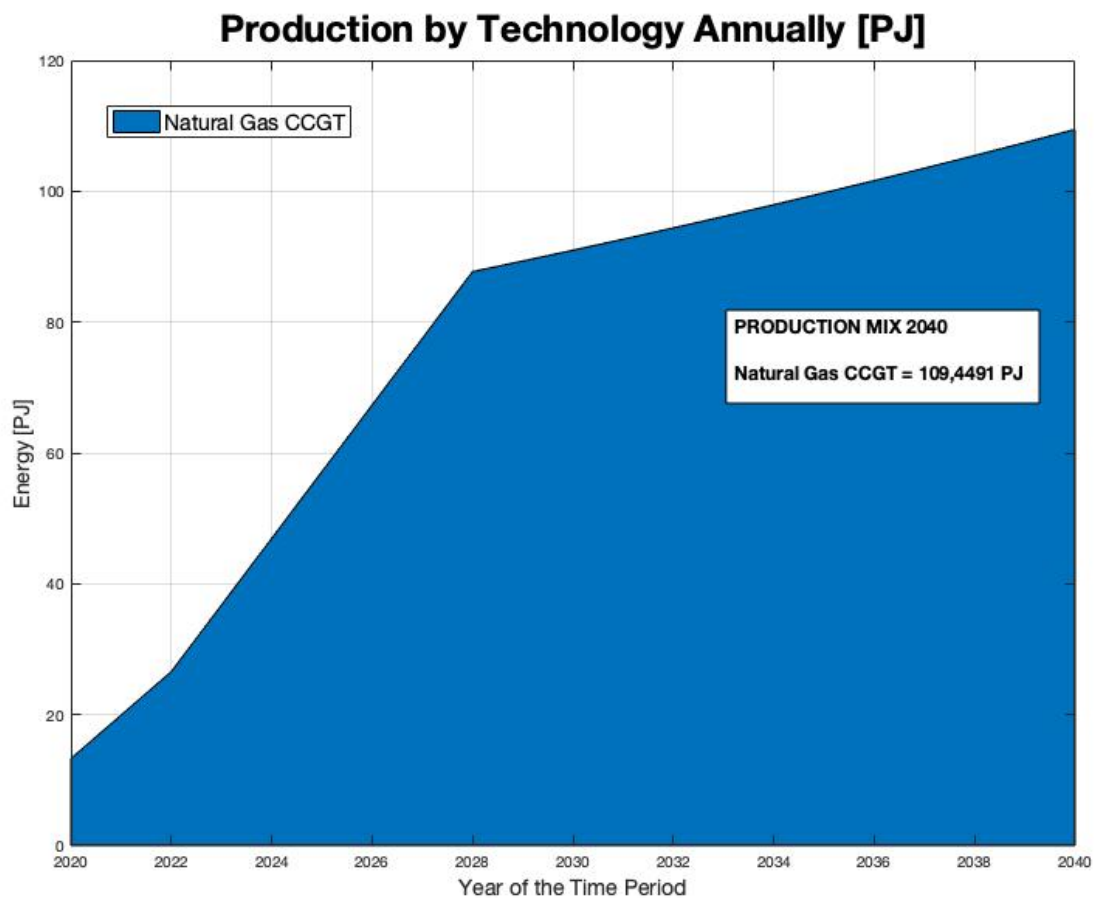


Figure 20. Production of electricity each year [PJ] by CCGT using natural gas.

Furthermore, the total discounted cost for satisfying the electricity demand and the correlating emissions can be seen in **Table 7**.

Table 7. Total discounted cost and total amount of emissions for the whole model period in the reference case

Variable	Reference case
Total Discounted Cost for whole model period [\$ billion]	10,3448
Total Amount of Emissions for whole model period [MTon CO ₂ -eq]	263,1

9.2 Scenario 1 - Nuclear Power and Natural Gas

The first scenario’s production mix consisted of both natural gas and nuclear, presented in **Figure 21**. Nuclear was introduced in 2026 with the completion of the first nuclear reactor, phasing out natural gas two years later at 2028 as the three remaining reactors are finished. Nuclear covers the vast majority of the remaining production, with a small amount of natural gas being reintroduced 2036. As shown in **Figure 22**, this was one year after nuclear reached its maximum allowed capacity of 4,8 GW.

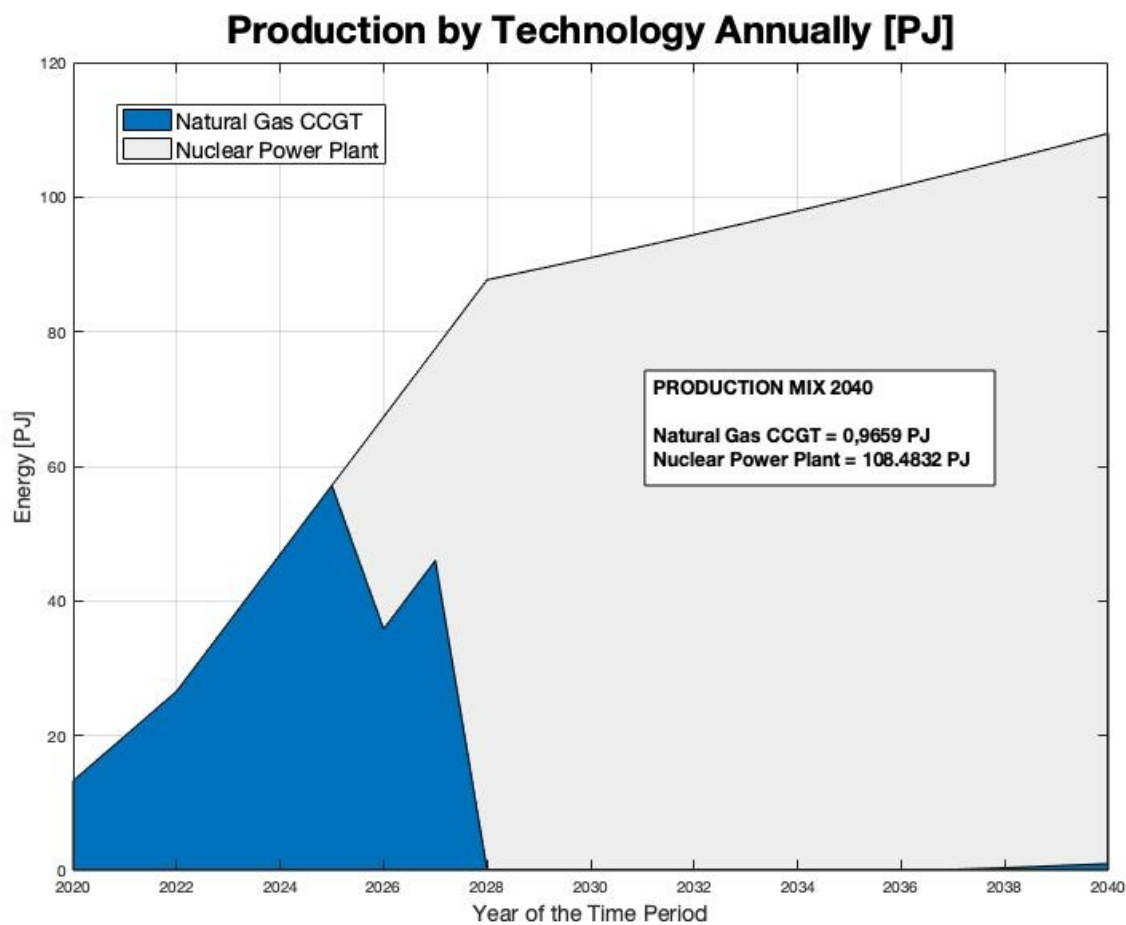


Figure 21. Production of electricity each year [PJ] by available technologies in scenario 1.

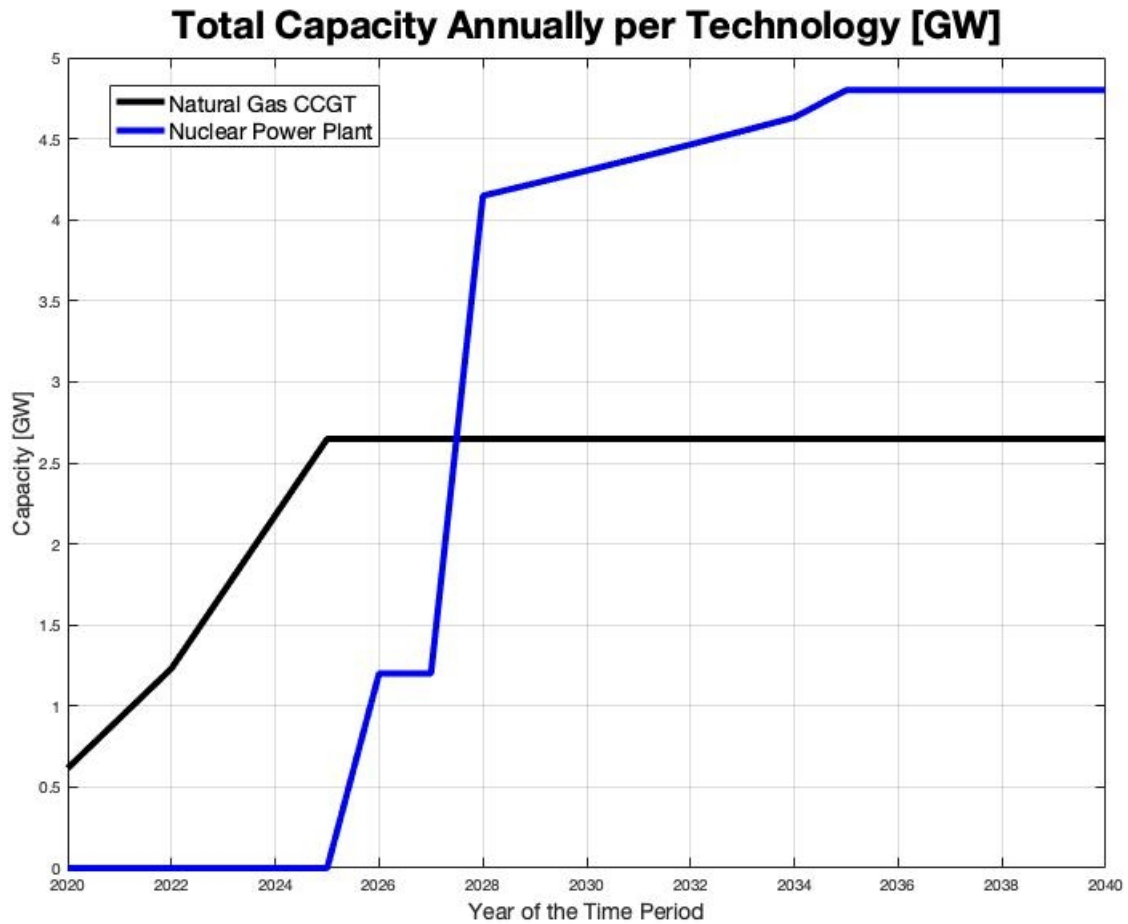


Figure 22. Total installed capacity each year for all available technologies in scenario 1.

Furthermore, the total discounted cost for satisfying the electricity demand and the correlating emissions can be seen in **Table 8**.

Table 8. Total discounted cost and total amount of emissions for the whole model period in scenario 1

Variable	Scenario 1
Total Discounted Cost for whole model period [\$ billion]	20,4545
Total Amount of Emissions for whole model period [MTon CO ₂ -eq]	46,2

9.3 Scenario 2 – 50 % Renewables by 2040

The production mix for scenario 2 consisted of several technologies in 2040, which can be seen in **Figure 23** and **Figure 24**. Almost all renewable energy sources contributed to the mix, despite for CSP which appeared to be too expensive and was therefore not in the mix and had no capacity installed. Furthermore, it can be seen that the WtE plant increased its production throughout the whole model period, i.e. 2020 to 2040, since the population increased and the amount of waste available for electricity production increased as well. The WtE was set to produce at maximum rate at all times as a way of handling the waste in El Alamein, which also contributed to its increase. Natural gas stood for a majority of the electricity production during the whole time period except for year 2040, where it accounted for 50 %, which was the maximum allowed share that year. Lastly, it can be seen that solar PV was the single biggest contributor to the mix from renewables in 2040, followed by the second largest contributor, onshore wind power.

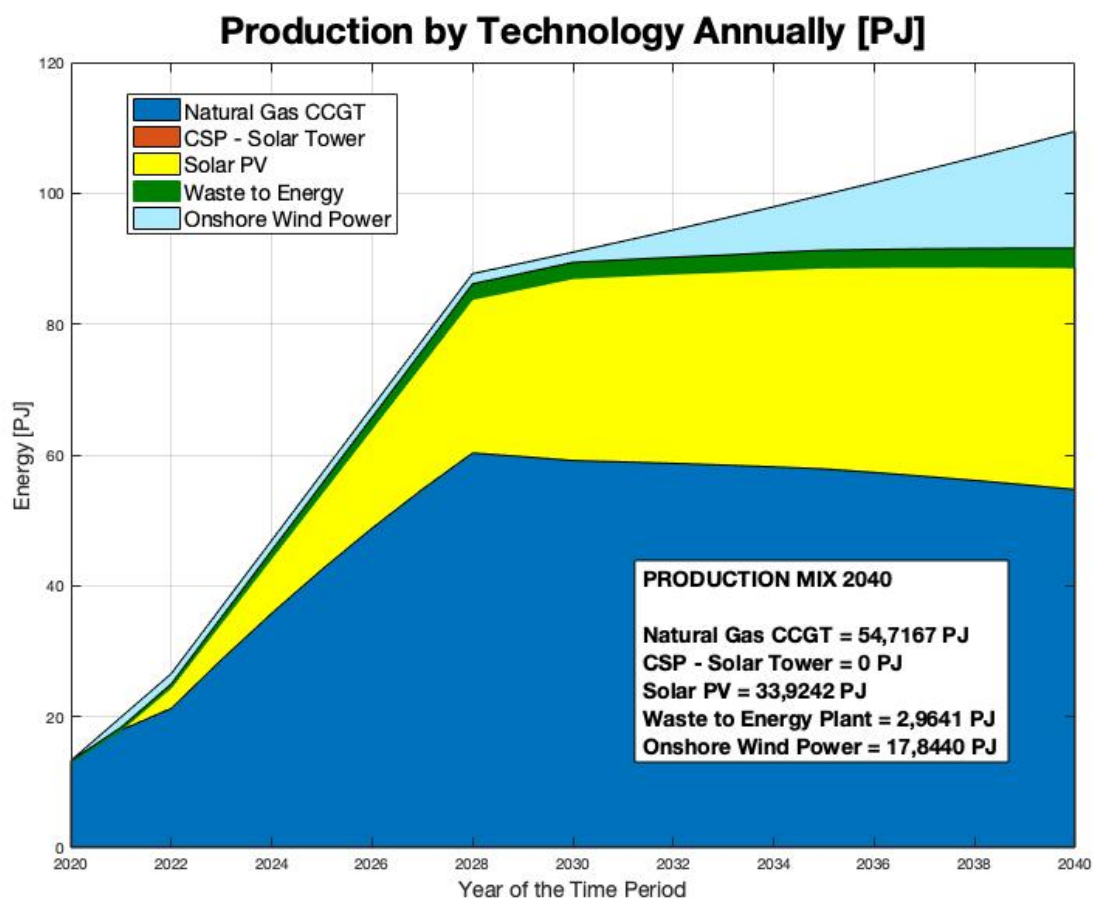


Figure 23. Production of electricity each year [PJ] by available technologies in scenario 2.

With the installed capacity of onshore wind power in 2040 as seen in **Figure 24**, and the area needed for one MW mentioned in the wind technology section i.e. 0,2186 km²/MW, the total area usage in 2040 amounted to about 433 km². Solar PV had its maximum capacity installed at all years after 2030, which can be seen in **Figure 24** if one compares it to Appendix 7.4, and the capacity was gradually growing as the available rooftop area increased.

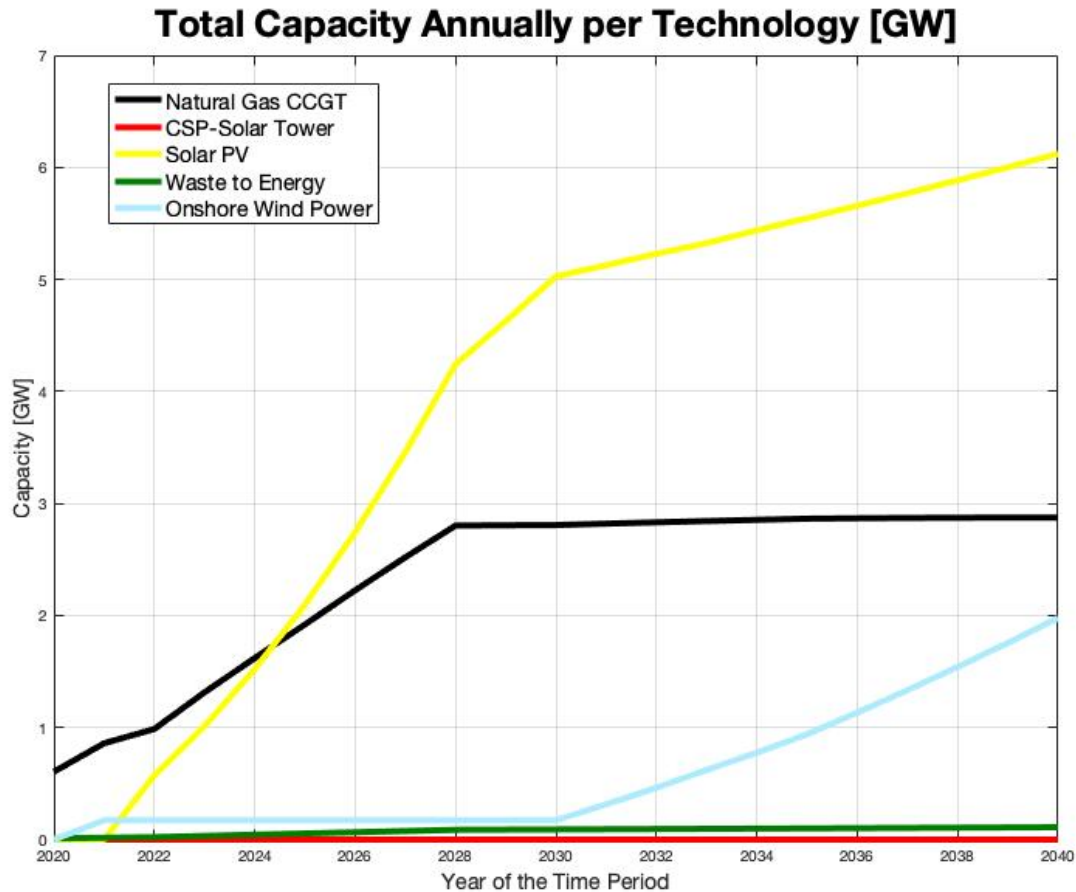


Figure 24. Total installed capacity each year for all available technologies in scenario 2.

Furthermore, the total discounted cost for satisfying the electricity demand and the correlating emissions can be seen in **Table 9**.

Table 9. Total discounted cost and total amount of emissions for the whole model period in scenario 2

Variable	Scenario 2
Total Discounted Cost for whole model period [\$ billion]	10,8603
Total Amount of Emissions for whole model period [MTon CO ₂ -eq]	164,6

9.4 Scenario 3 – 100 % Renewables by 2040

The electricity production in scenario 3 was dominated by the three technologies solar PV, onshore wind power and natural gas CCGT, as seen in **Figure 25**. Natural gas CCGT stood for a majority of the electricity production initially, reaching the peak of its production in 2027. The share of production from solar PV increased rapidly during this period, reaching its maximum allowed capacity in 2028. The technology continued to do so throughout the rest of the model period, gradually growing as the available rooftop area increased, see **Figure 25** and Appendix 7.3 for comparison. With the beginning in 2028, big investments were made into wind and going forward from 2033, the technology stood for the largest share of the electricity production, resulting in close to two thirds of the production in 2040.

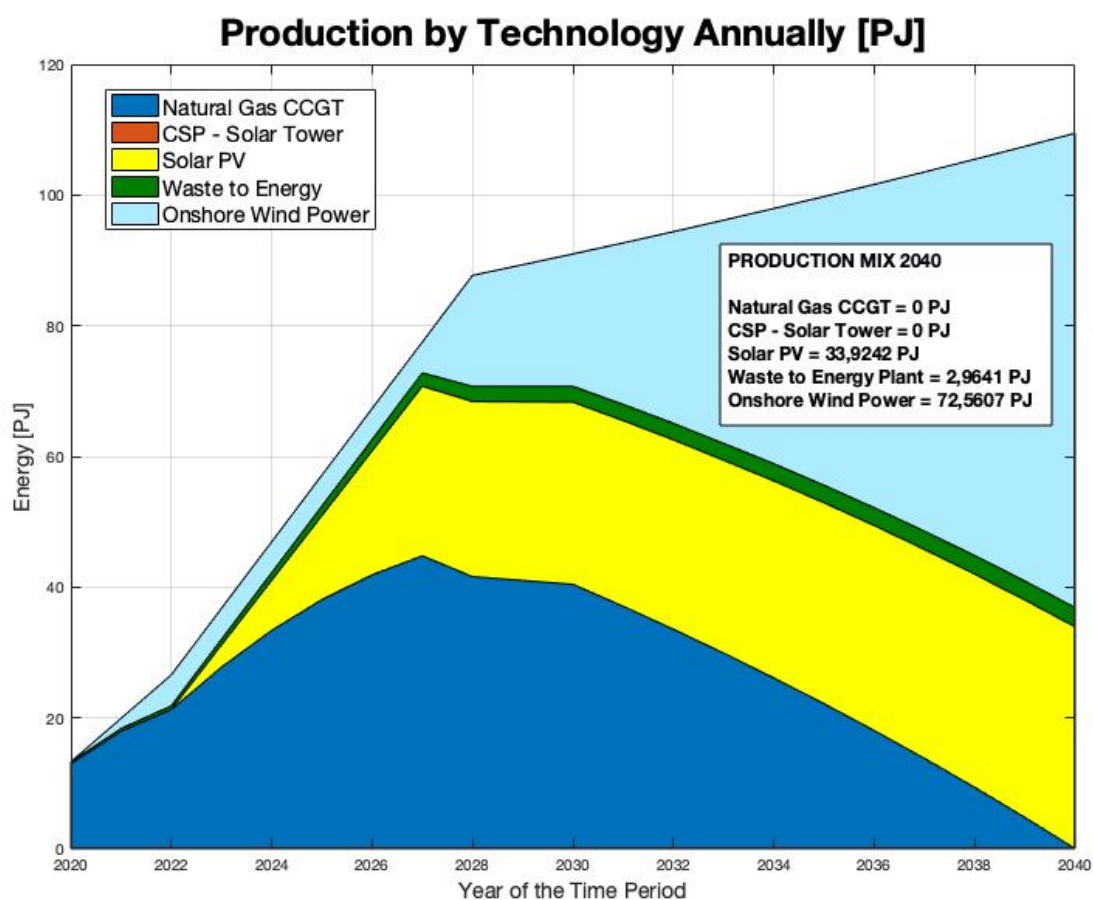


Figure 25. Production of electricity each year [PJ] by available technologies in scenario 3.

With the installed capacity of onshore wind power in 2040, as seen in **Figure 26**, and the previously mentioned area needed for one MW, the total area usage amounted to roughly 2705 km². Furthermore, as in scenario 2, CSP appeared to be too expensive and no investment was done into the technology. Moreover, the WtE technology acted identical as to the previous scenario.

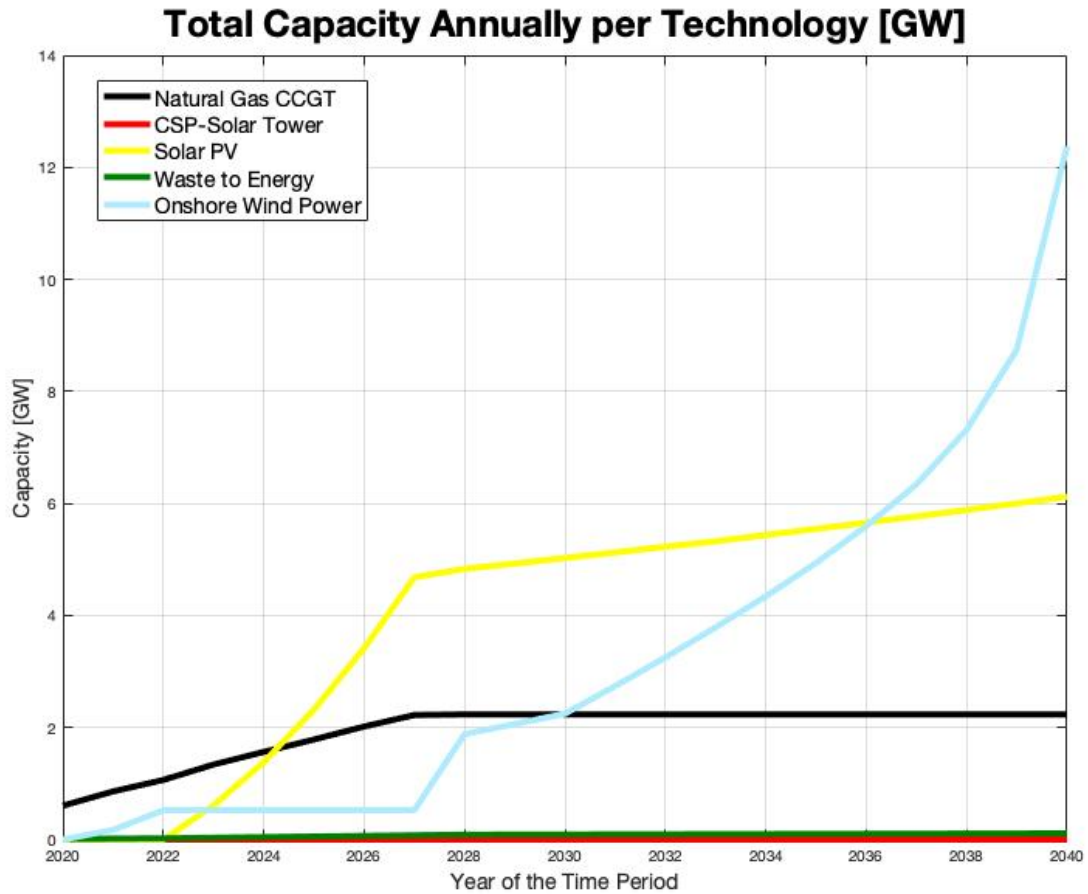


Figure 26. Total installed capacity each year for all available technologies in scenario 3.

Furthermore, the total discounted cost for satisfying the electricity demand and the correlating emissions can be seen in **Table 10**.

Table 10. Total discounted cost and total amount of emissions for the whole model period in scenario 3

Variable	Scenario 3
Total Discounted Cost for whole model period [\$ billion]	12,2107
Total Amount of Emissions for whole model period [MTon CO ₂ -eq]	90,3

9.5 Cost reduction of emissions

By comparing the difference in total discounted cost and amount of emissions of all scenarios to the reference case, the cost reduction of emissions for each case was identified and are presented in **Table 11**.

Table 11. Difference in variables between scenarios and the reference case

Comparison to Reference Case	Scenario 1	Scenario 2	Scenario 3
Difference in cost [\$ billion]	10,1097	0,5155	1,8659
Difference in emissions [Mton CO ₂ -eq]	216,9	98,5	172,9
Cost reduction per emissions [\$/ton CO ₂ -eq]	46,61	5,23	10,79

10. Discussion

In scenario 2, solar PV stood for the majority of the renewable electricity production whereas in scenario 3, onshore wind power stood for the majority, see **Figure 23** and **Figure 25**. Moreover, in scenario 2 solar PV reached its maximum allowed capacity in the year 2030, while in scenario 3 the technology reached its maximum allowed capacity two years earlier, i.e. 2028. In both cases the capacity remained at the highest allowed level moving forward, see **Figure 24**, **Figure 26** and Appendix 7.4. The WtE production remained the same in both scenarios, which is expected since its production was predetermined. Natural gas increased in the beginning of both scenarios and declined in line with each renewable production target, resulting in a bigger decline in scenario 3 than in scenario 2. CSP did not make an appearance in any of the two renewable scenarios.

The difference in total discounted cost between both renewable scenarios, i.e. scenario 2 and scenario 3, was around \$1,35 billion in favour of scenario 2, see **Table 9** and **Table 10**, although the amount of CO₂-eq emissions were higher in the second scenario, 164,6 Mton compared to 90,2 Mton. The most cost-efficient solution in meeting the demand of all scenarios was the one based on the current energy situation where natural gas was exclusively utilized in CCGT plants, closely followed by scenario 2. It was also the scenario which produced the most CO₂-eq, see **Table 11**. The least cost-efficient scenario in meeting the demand was the one based on Egypt's solution, containing a mixture of natural gas and nuclear. This was also the scenario which produced the least amount of CO₂-eq emissions, see **Table 11**.

Comparing scenario 2 and the reference case, the main difference between the two scenarios was the large contribution of solar PV and onshore wind power. The reference case's total discounted cost was \$0,5155 billion lower than that of scenario 2, and released 98,5 Mton more CO₂-eq. A possible reason for the renewable technologies in scenario 2 being more expensive, is the fact that the CCGT technology has been practiced for a longer period of time and has therefore had more time to be optimized. Since the technology solar PV had a limited maximum allowed capacity, this comparison is not able to determine whether it is more cost efficient compared to natural gas. With that said, the fact that both scenario 2 and 3 reached its maximum capacity of solar PV early on, indicates that the technology is more preferable compared to the other renewable technologies. Therefore, if solar PV farms were allowed within the model, the share of solar PV would likely increase. The result demonstrated that scenario 2 was more optimal from an environmental point of view, but not from an economic point of view, which might be the reason why it has not been implemented as a scenario in real life for El Alamein.

While all scenarios met the same demand, they did so with different costs and produced different amounts of emissions. Therefore, the increased costs in relation to lower emissions, put a price on the reduction of emissions. Scenario 2 required a higher total discounted cost compared to the reference case, while it emitted a lower amount of CO₂-eq. By comparing the differences of both these variables between the two scenarios, the cost of reducing the difference in emissions resulted in \$5,23/ton CO₂-eq. Furthermore, the reduction cost obtained when comparing reference case with scenario 1, where nuclear power is added to the mix, resulted in \$46,61/ton CO₂-eq.

For the values presented above, see **Table 11**. It is therefore evident that scenario 2 provides a much more cost-efficient solution, almost nine times as efficient, for reducing emissions compared to the solution chosen by Egypt.

As for scenario 3, the technologies behaved the same as they did in scenario 2, where investments were made into onshore wind power after maximising the solar PV capacity. Onshore wind power was the most used technology, waste was producing at maximum capacity, which was applied to solar PV as well for the years after 2028. Another similarity between the scenarios was the absence of CSP. Even though the cost trend for the technology was quite significant in its decline, it appeared not to be enough. The result was that scenario 3 had the same outcome as scenario 2, where it reduces emissions, but the energy system was more expensive than that of the reference case with natural gas. As for the cost comparison with the reference case, the cost reduction amounted to \$10,79/ton CO₂-eq, see **Table 11**. This equals a lower cost reduction than Egypt's solution with nuclear power but not that of scenario 2, which remains the most cost-efficient solution by a substantial amount, for reducing emissions.

As previously mentioned, there are several similarities between scenario 2 and scenario 3. With that said, and as mentioned above, scenario 2 with 50 % renewable energy provided the most cost-efficient solution when taking the emissions and the total discounted cost into account. This means that if Egypt wants to reduce its emissions by replacing natural gas with renewables, then scenario 2 should be chosen between the two renewable scenarios. Assuming there is a limited budget with the purpose of reducing emissions, the money that would be saved by choosing the less expensive alternative could then be used to reduce emissions in a more cost-efficient way at some other location, if one also assumes the same prerequisites. Furthermore, both scenarios with renewables proved to be more cost efficient at meeting the electricity demand as well as reducing emissions compared to scenario 1, Egypt's solution with nuclear power. Finally, if Egypt would have introduced a tax or penalty on carbon dioxide emissions, greater than the reduction cost for scenario 2, the latter alternative would have then been the most cost-efficient scenario.

Considering different renewable energy technologies, solar PV proved to be the most cost efficient, and since it stood for the biggest share of the renewable electricity production in scenario 2, it is the reason for the scenario being the most cost efficient between the two renewable ones. The fact that solar PV's implementation was limited to the amount of rooftop area available, makes it highly likely that this limited its contribution to the electricity mix in both renewable scenarios. Therefore, if installation was allowed in other forms such as large-scale PV farms, it would be expected to have an increased share in the electricity mix. However, since the production from solar PV is dependent on the availability of sunlight, it cannot single handily satisfy the entire demand for one day. Onshore wind power would therefore be a good complement, as its production does not vary between night and day.

The capacity needed would also require an area, but in our model, it was not a limiting factor for onshore wind power as it was for solar PV. In reality, an area of 433 km² would be needed for the 2 GW wind power that was installed by 2040 in scenario 2, which is almost 50 % bigger than the entire city. In scenario 3, the area needed to implement the capacity of wind power in 2040 amounted to 2705 km², which equals a size nine times larger than the city, something that

is arguably unreasonable to implement in reality. One reason for it being unreasonable might be that there are several other possible usages for the area that may be more advantageous. This was something our model did not take into consideration but is still something that absolutely affects the decision regarding the design of the energy system.

The two remaining renewable technologies, CSP and WtE, did not display any potential to stand for a large share of the electricity mix. Both technologies do however contain a few advantages. The CSP for example, is not as dependent on varying weather conditions in comparison to wind power and solar PV, due to its storage. This aspect was diminished in the model, since the time slices were not chosen based on the availability of sunlight, instead it was based on the demand profile. The advantage of storage was not as effective as it should have been, since all technologies could produce electricity during all time slices. As for WtE, our model indicates that it does not have a big potential for electricity production. A reason, for instance, was the lack of fuel, even though the assumption of 100 % food collection was a vastly unrealistic one. Although the assumption made on separation of organic matter from remaining municipal solid waste is unrealistic as well, it is not deemed likely to impact the electricity production drastically in relation to other technologies. It does however provide the service of waste management which otherwise would have been needed elsewhere. Also, the city of El Alamein will not have any agriculture which could have boosted the production rate for the WtE technology because of the increase in available fuel.

Since the model was based on a number of limitations, the results might not reflect reality. For instance, the project has used an approach characterized by a broad perspective. This has, as mentioned in the beginning of the report, influenced the results. This approach has, for example, affected the information gathered about the technologies. The cost trends for every technology have been assumed to follow historical trends. However, this was an oversimplification and may not be realistic. In some cases, these trends may have been too optimistic and could have benefited the specific technology concerned. For example, the cost trend for solar PV was based on a historical decline of 32 % in 4 years, which can be interpreted as a rather large decline and could indicate that the technology's cost trend may soon stagnate. Furthermore, this broad perspective contributed to the limitation of only considering electricity as a source of energy, and not heat. If heat would have been taken into account, it could have changed the overall performance of different technologies since their potential may not have been fully utilized.

Additionally, the process of physically implementing the different technologies were not taken into consideration. This includes problems such as public opinion regarding solar PV on every rooftop and the need for all residents to dispose their waste in a manner that makes it retrievable, for example. Finally, the choice of only letting the main fuel, natural gas, represent Egypt's current electricity production, affected the comparison between the different scenarios, resulting in a somewhat inaccurate observation. For example, the share of renewables as well as oil in the current energy mix is not taken into consideration which affects the comparisons between the different scenarios.

This project aimed to provide further knowledge regarding the necessary transition towards a sustainable energy solution in order to mitigate the effects of the ongoing climate change, arguably human kinds biggest challenge. Since this subject is of such great importance and relevance, it strengthens the significance of this report. Clean and affordable energy, also known as SDG 7, is one of the necessary solutions for this global challenge. This project provides the most cost-efficient approach to limit emissions while satisfying the electricity demand in the city of El Alamein. Since this approach is characterized by low emissions, it is also in line with SDG 13, climate action. Moreover, because of Egypt's own goals regarding the fulfilling of the SDGs as well as their national goals of increasing the amount of electricity provided by renewable energy sources, this project provides insights for the possible establishment of new policies and actions. Policies and actions needed for the transition towards a future defined by sustainability.

11. Conclusion and future work

Firstly, there are four identified renewable energy technologies capable of supplying electricity at the identified demand to the city of El Alamein, where solar PV established itself as the most cost-efficient renewable technology. Secondly, of all the scenarios compared to the reference case, scenario 2 with 50 % renewables proved to be the most cost-efficient energy mix for meeting the defined demand. This scenario also proved the most cost efficient at reducing emissions by a large margin. The above conclusions culminate into great optimism regarding the future use of renewable energy in Egypt, with the solution for El Alamein as a source of inspiration.

Although our project achieved its purpose, with new knowledge obtained, new questions and problems arose. For future studies there are several aspects not highlighted in this report that are worth considering. For example, taking the area needed for each technology into account in the model as well as the actual implementation process, since potential problems surrounding these aspects have not been thoroughly examined. Furthermore, studies regarding the technologies that did not prove fruitful in our model would be encouraged in other areas, where other prerequisites might allow different results, e.g. places where agriculture provides crop residue for the waste technology. Examining the prospects of other types of technologies such as large-scale solar PV farms and wave technology, would provide useful information for future energy system analysis. Finally, a more detailed model of El Alamein's energy system with input data more specific for the area would be advantageous since this might provide a more accurate image of cost for each technology. More detailed time slices would also be preferred to provide a more detailed model.

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Appendices

1. Transcript from interview in El Alamein

Interview with the Electrical Manager Eng. Ahmed Hassan and Eng. Abd El Hamid

Q: Question, A: Answer

- **Q: The population of each phase and the total population?**
A: 1 million residents in phase 1, four phases so total population will be 4 million.
- **Q: Any assumption of the total electricity demand for the city?**
A: Phase 1 will have a demand of 750 MW, phase 2: 1325 MW. No information for phase 3 and 4.
- **Q: Any plans for renewable energy technologies in the city?**
A: Solar PV was planned from the beginning, but since the national grid is so available, we scrapped the idea of renewables.
- **Q: What kind of industry will operate in the city?**
A: Mainly petrochemical industry and a recycle plant. The recycle plant has not decided what materials it will recycle. There is still room for industry to be built, although we have not decided what kind.
- **Q: When do you expect the El Daaba nuclear power plant to be finished and will you use it?**
A: The grid to Daaba is already built, the first phase of the nuclear plant will be done 2026 and we expect it to supply us from the national grid.
- **Q: When are the different phases done?**
A: First phase started March in 2018 and will be done June 2020. No information on the other phases.
- **Q: What does every phase involve?**
A: First phase will involve universities, residential and tourism. When investments come from tourism, the rest of the city will expand.
- **Q: Are you planning for any agriculture?**
A: No agriculture.
- **Q: Water demand, the method of purification and the electricity demand to purify?**
A: The method will be reverse osmosis. There will be total of 3 water purifications stations using water from the sea, each station will produce 150 000 m³/day and has an electricity demand of 30 MW. That's a total of 450 000 m³/day and electricity demand of 90 MW.
- **Q: Any calculations on the city's predicted CO₂-emissions?**
A: No information available.
- **Q: Have you looked at wave technology?**
A: The beaches will be used for tourism, if it is to be installed it has to be on the outskirts of the city's coast.
- **Q: Will there be any type of collective transportation available?**
A: Plans for metro, trams and busses. A railway will be built from El Alamein to Alexandria and Cairo.

2. Onshore Wind Power

2.1 Capital cost

Capital cost each year for Onshore Wind Power [\$/MW]

2020	1 200 000	2027	936 547,4	2034	730 934,1
2021	1 158 250,4	2028	903 963,6	2035	705 503,9
2022	1 117 953,3	2029	872 513,5	2036	680 958,5
2023	1 079 058,2	2030	842 157,6	2037	657 267,1
2024	1 041 516,3	2031	812 857,8	2038	634 399,9
2025	1 005 280,6	2032	784 577,4	2039	612 328,3
2026	970 305,6	2033	757 280,9	2040	591 024,5

2.2 Variable O&M cost

Variable O&M cost each year for Onshore Wind Power [\$/MWh]

2020	20	2027	15,6091	2034	12,1822
2021	19,3042	2028	15,0660	2035	11,7584
2022	18,6326	2029	14,5419	2036	11,3493
2023	17,9843	2030	14,0360	2037	10,9545
2024	17,3586	2031	13,5476	2038	10,5733
2025	16,7547	2032	13,0763	2039	10,2055
2026	16,1718	2033	12,6213	2040	9,8504

2.3 Capacity factors

Capacity Factor – Time Slice

	Day	Night
January	0,3089	0,3089
February	0,2456	0,2456
March	0,3378	0,3378
April	0,2890	0,2890
May	0,2659	0,2659
June	0,2319	0,2319
July	0,2617	0,2617
August	0,2189	0,2189
September	0,2336	0,2336
October	0,2994	0,2994
November	0,2086	0,2086
December	0,2870	0,2870

Adjusted Capacity Factor – Time Slice

	Day	Night
January	0,3317	0,3317
February	0,2637	0,2637
March	0,3627	0,3627
April	0,3103	0,3103
May	0,2855	0,2855
June	0,2490	0,2490
July	0,2810	0,2810
August	0,2350	0,2350
September	0,2508	0,2508
October	0,3215	0,3215
November	0,2240	0,2240
December	0,3082	0,3082

3. Solar PV

3.1 Capital cost

Capital cost each year for Solar PV [\$/MW]					
2020	1 200 000	2027	611 107	2034	311 209
2021	1 089 720	2028	554 946	2035	282 609
2022	989 575	2029	504 946	2036	256 638
2023	898 633	2030	457 634	2037	233 053
2024	816 049	2031	415 577	2038	211 635
2025	741 054	2032	377 386	2039	192 186
2026	672 951	2033	342 704	2040	174 524

3.2 Capacity factors

Average monthly Capacity Factor for Solar PV			
Jan	13,56 %	Jul	19,39 %
Feb	17,43 %	Aug	19,88 %
Mar	18,37 %	Sep	19,63 %
Apr	19,94 %	Oct	18,67 %
May	18,48 %	Nov	13,30 %
Jun	19,10 %	Dec	13,06 %

Adjusted Capacity Factor – Time Slice		
	Day	Night
January	0,1830	0,0882
February	0,2299	0,1187
March	0,2435	0,1239
April	0,2641	0,1347
May	0,2423	0,1273
June	0,2498	0,1322
July	0,2534	0,1344
August	0,2597	0,1379
September	0,2565	0,1361
October	0,2446	0,1288
November	0,1745	0,0915
December	0,1718	0,0894

4. CSP – Solar Tower

4.1 Capital cost

Capital cost each year for CSP – Solar Tower [\$/MW]

2020	7 880 000	2027	5 316 657	2034	3 587 162
2021	7 449 270	2028	5 026 042	2035	3 391 084
2022	7 042 084	2029	4 751 312	2036	3 205 723
2023	6 657 155	2030	4 491 600	2037	3 030 494
2024	6 293 267	2031	4 246 084	2038	2 864 844
2025	5 949 270	2032	4 013 988	2039	2 708 248
2026	5 624 075	2033	3 794 578	2040	2 560 212

4.2 Variable O&M cost

Variable O&M cost each year for CSP – Solar Tower [\$/MWh]

2020	31	2027	20,916	2034	14,112
2021	29,306	2028	19,772	2035	13,341
2022	27,704	2029	18,692	2036	12,611
2023	26,189	2030	17,670	2037	11,922
2024	24,758	2031	16,704	2038	11,270
2025	23,404	2032	15,791	2039	10,654
2026	22,125	2033	14,928	2040	10,072

4.3 Capacity factors

Average monthly Capacity Factor for CSP – Solar Tower

Jan	50,3119 %	Jul	59,0845 %
Feb	55,1431 %	Aug	60 %
Mar	56,5847 %	Sep	59,4990 %
Apr	58,9755 %	Oct	57,7586 %
May	57,3188 %	Nov	50,4561 %
Jun	58,4629 %	Dec	50 %

Adjusted Capacity Factor – Time Slice

	Day	Night
January	0,6678	0,3286
February	0,7428	0,3600
March	0,7622	0,3694
April	0,7944	0,3852
May	0,7720	0,3742
June	0,7874	0,3818
July	0,7958	0,3858
August	0,8082	0,3918
September	0,8014	0,3886
October	0,7780	0,3772
November	0,6798	0,3296
December	0,6734	0,3266

5. Nuclear Power plant (BAU)

5.1 Variable O&M cost

Variable O&M cost each year for Nuclear Power plant [\$/MWh]					
2026	27,3	2031	21,8454	2036	17,4806
2027	26,1097	2032	20,8929	2037	16,7185
2028	24,9713	2033	19,9820	2038	15,9895
2029	23,8825	2034	19,1108	2039	15,2924
2030	22,8413	2035	18,2776	2040	14,6256

6. Natural Gas CCGT (BAU)

6.1 Variable O&M cost

Variable O&M cost each year for Natural Gas CCGT [\$/MWh]					
2020	21,650	2027	25,861	2034	31,018
2021	22,201	2028	26,535	2035	31,843
2022	22,767	2029	27,229	2036	32,693
2023	23,351	2030	27,944	2037	33,568
2024	23,951	2031	28,679	2038	34,369
2025	24,570	2032	29,436	2039	35,396
2026	25,206	2033	30,215	2040	36,350

7. Preparations for OSeMOSYS

7.1 Specified annual demand

Specified annual demand [fraction of total demand]		
	Day	Night
January	0,0400	0,0357
February	0,0372	0,0333
March	0,0397	0,0355
April	0,0406	0,0363
May	0,0450	0,0363
June	0,0496	0,0443
July	0,0517	0,0461
August	0,0528	0,0471
September	0,0471	0,0421
October	0,0447	0,0399
November	0,0394	0,0351
December	0,0406	0,0362

7.2 Electricity demand in El Alamein

Electricity demand in El Alamein each year [PJ]					
2020	11,8260	2027	68,9850	2034	87,1540
2021	17,7390	2028	78,0516	2035	88,7790
2022	23,6520	2029	79,4950	2036	90,4360
2023	32,7186	2030	80,9670	2037	92,1260
2024	41,7852	2031	82,4690	2038	93,8490
2025	50,8518	2032	84,0000	2039	95,6070
2026	59,9184	2033	85,5620	2040	97,4000

7.3 Available rooftop area for Solar PV

Total available rooftop area each year in El Alamein [m ²]					
2020	6 040 000	2027	24 160 000	2034	27 192 083
2021	6 040 000	2028	21 160 000	2035	27 733 206
2022	12 080 000	2029	24 640 784	2036	28 285 096
2023	12 080 000	2030	25 131 136	2037	28 847 970
2024	18 120 000	2031	25 631 245	2038	29 422 044
2025	18 120 000	2032	26 141 307	2039	30 007 543
2026	24 160 000	2033	26 615 191	2040	30 604 693

7.4 Maximum Solar PV capacity

Maximum Solar PV capacity each year in El Alamein [GW]

2020	1,208	2027	4,832	2034	5,438
2021	1,208	2028	4,832	2035	5,546
2022	2,416	2029	4,928	2036	5,657
2023	2,416	2030	5,026	2037	5,769
2024	3,624	2031	5,126	2038	5,884
2025	3,624	2032	5,228	2039	6,001
2026	4,832	2033	5,323	2040	6,121

7.5 Total Organic Waste generated

Total Organic Waste Generated each year [Mton]

2020	0,08176	2027	0,57232	2034	0,73616713
2021	0,12264	2028	0,65408	2035	0,75081685
2022	0,16352	2029	0,66709619	2036	0,76575811
2023	0,24528	2030	0,68037141	2037	0,78099669
2024	0,32704	2031	0,6939108	2038	0,79653853
2025	0,4088	2032	0,70771962	2039	0,81238965
2026	0,49056	2033	0,72180324	2040	0,8285562

7.6 Electricity production from Organic Waste

Electricity Production from Organic Waste each year [PJ]

2020	0,2924964	2027	2,0474748	2034	2,6336379
2021	0,4387446	2028	2,3399712	2035	2,68604729
2022	0,5849928	2029	2,38653663	2036	2,73949963
2023	0,8774892	2030	2,43402871	2037	2,79401568
2024	1,1699856	2031	2,48246588	2038	2,84961659
2025	1,462482	2032	2,53186695	2039	2,90632396
2026	1,7549784	2033	2,5822511	2040	2,9641598

TRITA-ABE-MBT-19408